Bank Credit Supply and the Unemployment Rate*

Cory Langlais†
Mount Allison University

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Abstract

There is a strong negative relationship between U.S. bank credit and the unemployment rate since the mid-1980s. In this paper, I ask: *Is there a causal connection between changes in the supply of bank credit and the observed fluctuations of the unemployment rate?* If so, what are the transmission mechanisms governing such a relationship? To answer these questions, I proceed down two paths. First, I use a bank credit supply measure estimated by Bassett, Chosak, Driscoll, and Zakrajšek (2014) to document the effect that bank credit supply shocks have on the unemployment rate in a VAR model. The results suggest that a 1 percent contraction in the supply of bank credit leads to a 0.25 percentage point increase in the unemployment rate. Moreover, bank credit supply shocks account for more than one-third of the volatility in the unemployment rate. Second, I rationalize these results by incorporating a banking sector into an otherwise standard DSGE model with labor search frictions and nominal rigidities. Unlike standard banking models, this paper allows banks to finance loans through deposit (i.e., money) creation. Matched firms must obtain a bank loan to purchase their inputs before production. When banks contract credit, there are fewer funds for firms to purchase their capital and labor inputs and the cost of borrowing rises. Firms' benefit of posting a vacancy falls, leading to a decrease in labor market tightness. In both empirical and model settings, a contraction in bank credit supply leads to a typical recession: GDP, investment, and labor market tightness fall while the cost of external finance and the unemployment rate rise.
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Keywords: financing through money creation, deposit creation, SAM frictions, unemployment, New Keynesian DSGE model

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†Contact: Department of Economics, Mount Allison University, 62 York Street, Sackville, New Brunswick, Canada. Send comments to colanglais@mta.ca.

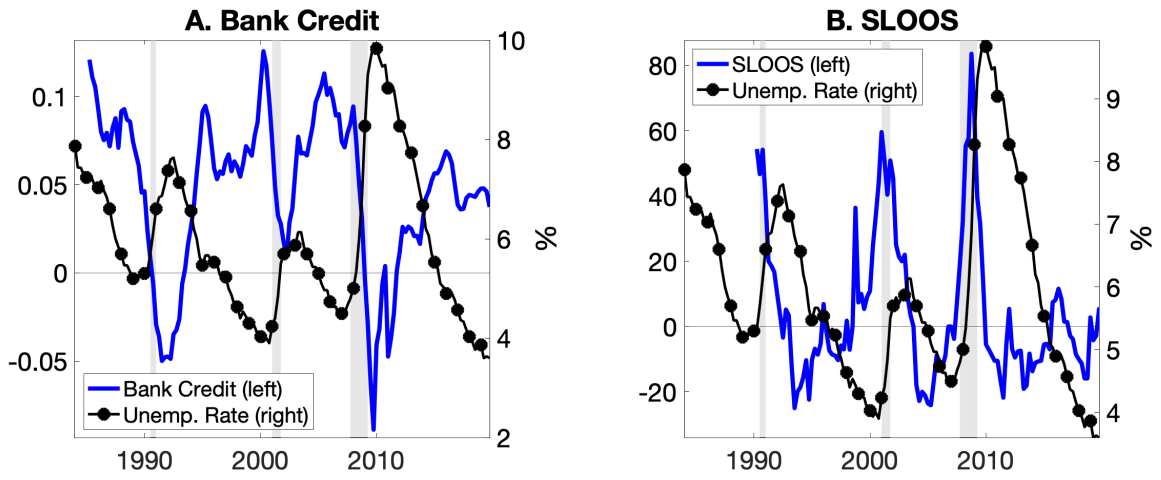

Figure 1: Bank Credit & Unemployment Rate

Notes: Shaded bars denote NBER recessions. Senior loan officer opinion survey (SLOOS) denotes net percentage of domestic banks tightening standards for C&I loans to large and middle-market Firms (DRTSCILM)
Sources: Bank for International Settlements; Bureau of Labor Statistics; Federal Reserve Board.

1 Introduction

Bank credit has long been understood to play an important role in the economy over the business cycle (see, e.g., Mises (1912) and Fisher (1933)). Recent historical papers find that financial market crises can predict and amplify economic downturns (Schularick and Taylor, 2012; Jordà, Schularick, and Taylor, 2013), while others have shown that disruptions to credit markets during the Great Depression and Great Recession contributed to the loss of employment (Chodorow-Reich, 2014; Benmelech, Frydman, and Papanikolaou, 2019). When there is an impending recession and growing uncertainty, firms cut investment spending while banks tighten lending standards leading to a reduction in bank-intermediated credit. Funds that would have otherwise been used to finance firm investment projects, for example, are not brought into circulation when bank lending is halted. Figure 1 captures the strong negative relationship between U.S. bank credit growth and the unemployment rate (panel A).[1,2] Each series exhibits a strong cyclical pattern as recessionary episodes (shaded areas) are associated with contracting credit and a rising unemployment rate. Of

[1] See Keen (2011) for a discussion regarding the relationship between the nominal/financial side and the real/goods side of the economy. Zanetti (2019) also discusses the relationship of firm debt and unemployment over the business cycle.

[2] I use the terms *bank credit* and *bank loans* interchangeably throughout the paper. Also, I use the term *bank* to refer to a deposit taking and loan granting financial institution that has been chartered by state or federal government and holds a reserve account with the Federal Reserve System.

course, the cyclicality of bank credit can be driven by changes in either supply or demand—or both. Moreover, in a world with financial frictions—giving rise to an external finance premium—changes in economic uncertainty or risk have been shown to be important determinants of intermediated credit (Gilchrist and Zakrajšek, 2012b; Christiano, Motto, and Rostagno, 2014). It is, therefore, not surprising that periods of substantial net tightening of bank lending standards (panel B, blue line) are associated with recessionary periods and a subsequent rise in the unemployment rate.[3]

I address the following questions in this paper: Do changes in the supply of bank credit contribute to the observed cyclical pattern of the unemployment rate? If so, how important are bank credit supply shocks at generating fluctuations of the unemployment rate? What are the transmission mechanisms governing such a relationship?

To answer these questions, I proceed down two paths. First, I conduct an empirical analysis using a measure of U.S. bank lending standards from Bassett et al. (2014) that is derived from bank-level senior loan officer opinion survey responses. The bank lending standards estimate is purged of factors that are known to influence loan demand, making it a plausible instrument of exogenous shifts in the bank credit supply curve. This purged lending standards measure is then included in a monetary VAR model to quantify the importance that exogenous changes in the supply of bank credit have on the unemployment rate. To the best of my knowledge, this is the first paper to explore the influence of U.S. bank lending standards on labor market outcomes. Second, to shed light on the transmission mechanism I introduce a banking sector that is consistent with the *financing through money creation* (FMC) view of banks into an otherwise standard labor search-and-matching (SAM) DSGE model with nominal frictions.[4] Unlike the standard banking models or lender/borrower relationships used in most papers (e.g., Petrosky-Nadeau (2014) and Monacelli, Quadrini, and Trigari (2023)), the FMC view assumes that banks finance their lending activity—in part or in whole—with newly created nominal bank deposits. As a result, if a bank wants to issue a loan to a firm, for example, the bank does not have to wait until a saver deposits a pre-accumulated amount of funds with the bank before the bank can issue the loan. Put differently, the bank can adjust its loans by creating additional deposits (i.e., money), thereby expanding the money supply.[5]

To the best of my knowledge, this is the first paper to model banks according to the FMC view in a SAM framework. As a result, my paper differs from the existing literature in three important ways. First, this paper allows the banking sector—under nominal rigidities—to be a potential source of economic fluctuations without relying on conventional borrowing or leverage constraints

[3]Bank lending standards denote the net percentage of banks reporting "tightening" their lending standards based on the *Senior Loan Officer Opinion Survey* (SLOOS).

[4]I borrow the term *financing through money creation* from Jakab and Kumhof (2015, 2019).

[5]Conversely, the bank also has the ability to curtail the amount of loans it offers, thereby contracting the amount of deposits (i.e., money) circulating in the economy when existing loans get repaid.

(e.g., Gertler and Kiyotaki (2010), Gertler and Karadi (2011), and Jermann and Quadrini (2012)). According to the senior loan officer opinion survey (SLOOS), the sharp reversal of commercial banks' lending standards during the financial crises was mainly due to changes in their economic outlook rather than changes in their capital or liquidity positions (Chen, Higgins, and Zha, 2021). This makes the banking sector of Gertler and Kiyotaki (2010) and Gertler and Karadi (2011) a poor candidate for studying the macroeconomic effects of bank credit supply shocks. Second, the banking sector in this paper is consistent with the description of commercial banks found in central bank publications (e.g., McLeay, Radia, and Thomas (2014) and Bundesbank (2017)) and with the empirical work of Werner (2014).[6] Third, the optimal lending behaviour of this paper's banking sector gives rise to an interest rate spread—i.e., an external finance premium—thereby allowing the credit supply channel to be disaggregated into a *price* and *quantity* effect.

The standard part of the model has labor search and nominal frictions, which closely follows Leduc and Liu (2016). Each period there are unemployed workers searching for jobs with firms who have chosen to post a vacancy. A matching technology brings a worker and firm together to create an employment relationship that lasts until an exogenous shock destroys the match. As mentioned above, the novel aspect of the model is the banking sector.[7] The banking sector accumulates net worth, offers deposit services to firms and households and issues loans to matched firms. All loans in the model are to the production sector.[8] Each period intermediate-good firms who are matched with a worker must obtain a bank loan to finance their wage and capital rental bills in advance of production. Each bank loan is financed using two sources of funds: newly created bank demand deposits and bank net worth. While bank net worth is a state variable and pre-accumulated household savings are governed by the household's Euler equation and not directly lent out to firms, the main margin of adjustment for bank loans are its demand deposits. As a result, I assume the bank is endowed with a proprietary FMC technology that allows it to adjust the level of FMC both endogenously and exogenously.[9] I refer to the exogenous adjustment of FMC as an 'FMC shock' and is meant to represent an exogenous bank credit supply shifter. FMC shocks can be thought of as changes in the bank's lending preferences based on factors not associated with their liquidity—i.e.,

[6]The unique function of commercial banks—differentiating them from other financial intermediaries—is nicely summarized by Mark Carney: "The principal way banks create money is by making a loan. When the bank decides a borrower is creditworthy [...] it credits their deposit account for the amount of the loan and new money enters circulation." (Carney, Mark, 2021, p. 60)

[7]As discussed below, there are two additional differences between this model and Leduc and Liu (2016) besides the banking sector. Namely, I add money in the form of currency and deposits—along with household liquidity preference—and I add capital stock to the model.

[8]While a nontrivial amount of U.S. commercial bank lending is to households in the form of consumer loans and residential mortgages, I assume there are no loans in the household sector. In the context of this paper, I believe this assumption to be warranted since it has been shown empirically that investment and corporate debt play a particularly important role in the transmission of financial frictions and credit disruptions over the business cycle (Etoundi Atenga, Abdo, and Mougoué, 2021; Ivashina, Kalemli-Özcan, Laeven, and Müller, 2024).

[9]The FMC technology used in this paper is a simplified version of that introduced in Langlais (2023a).

reserve—availability or capital—i.e., net worth—position.[10] In the benchmark model, the FMC shock process is treated as an unobservable and is, therefore, inferred from macroeconomic data used in the Bayesian estimation procedure.

The empirical results suggest that exogenous shifts in bank credit supply account for over one-third of the volatility in the vacancy and unemployment rate at business cycle frequencies. Moreover, a 1 percent contraction in the supply of bank credit leads to a 0.25 percentage point rise in the unemployment rate. In both the empirical VAR and estimated DSGE model, an unexpected contraction in the supply of bank credit leads to a typical recession: GDP, credit and labor market tightness fall while the cost of external finance and the unemployment rate rise. According to the model, when the bank unexpectedly contracts its supply of loans, matched firms experience a (*i*) shortage of funds required to purchase their inputs, and (*ii*) a rising cost of external finance. As a result, both the quantity effect (*i*) and price effect (*ii*) reinforce one another resulting in a fall in demand for capital and labor inputs. The lower demand for factor inputs results in lower investment and GDP. Furthermore, firms that are able to attain financing following the credit contraction face higher borrowing costs, which reduces the value of being matched with a worker. This leads to firms posting fewer vacancies and thereby reducing labor market tightness.

Since the FMC shock is treated as an unobservable, there is a possibility the estimated FMC shock may be capturing other sources of variation. To test whether this is the case, I perform two tests. First, I perform an out-of-sample test which compares the bank credit supply shock estimated in Bassett et al. (2014) with that estimated from the model. I find that the model estimated FMC shock does a reasonable job at capturing movements in the supply of bank credit especially in the period surrounding the 2007-09 financial crisis. For the second test, I employ an alternative identification method to estimate the model implied FMC shock. Compared to the benchmark FMC shock estimate, the alternative estimate is measured directly from bank balance sheet data. The results suggest the FMC shock played a crucial role during the 2007-09 financial crisis, but only a minor role during the 2001 recession.

The remainder of the paper is organized as follows. Section 2 provides a literature review. Section 3 describes the empirical bank credit supply shock used in the VAR model and analyzes the macroeconomic effects a bank credit supply shock has on the unemployment rate. Section 4 presents the DSGE model and performs equilibrium analyses. Section 5 describes the estimation procedure and presents the quantitative model results. Section 6 performs various out-of-sample tests. Section 7 concludes the paper.

[10]For example, exogenous changes in the bank's economic outlook, future uncertainty or industry competitiveness may be potential factors driving banks to either expand or contract their lending from one period to the next. See Chen et al. (2021) for a discussion regarding reasons bank loan officers give for changing their lending standards.

2 Related Literature

This paper sits at the intersection of two large bodies of literature: macro-finance literature and the labor SAM literature.

The macro-finance literature can be traced back to Carlstrom and Fuerst (1997) and Bernanke, Gertler, and Gilchrist (1999), which focused on information asymmetries in the lender-borrower relationship. Like my paper, these papers give rise to a financial friction embedded in the cost of external finance. There are, however, three key distinctions between my paper and traditional macro-finance literature. First, in my paper a key financial friction is a result of the bank's lending technology and not a result of agency costs between the lenders and borrowers. There are no information asymmetries between lenders and borrowers in my model as all loans are riskless intraperiod working capital loans. Second, my paper does not rely on leverage constraints (Gertler and Kiyotaki, 2010; Gertler and Karadi, 2011) or collateral constraints (Kiyotaki and Moore, 1997; Jermann and Quadrini, 2012) in order for the financial sector to contribute to, or amplify, aggregate fluctuations in real variables. Instead, the financial frictions in my model endogenously arise from the bank's optimality condition and lending technology. Third, unlike the early macro-finance literature which focuses on the propagation and amplification mechanisms of shocks originating outside the financial sector—such as monetary policy or various productivity shocks—the financial sector in my paper is a potential source of volatility—via the FMC shock.

This is not the first paper in which shocks originating in the lender/borrower relationship can play a role in generating economic fluctuations.[11] For example, Jermann and Quadrini (2012) incorporate a Kiyotaki and Moore (1997)-style collateral constraint subject to shocks on firm debt finance. They call these financial shocks. Like my paper, they find that financial shocks are important for both the financial and real sector, especially employment.[12] Extending the Bernanke et al. (1999) financial accelerator framework with nominal frictions, Christiano, Motto, and Rostagno (2003) and Nolan and Thoenissen (2009) estimate various structural shocks and find that shocks to a borrower's net worth are an important source of volatility in the U.S. economy. Goodfriend and McCallum (2007) and Jakab and Kumhof (2019) each develop a banking sector which is a potential source of volatility. Goodfriend and McCallum (2007), however, focus on monetary policy transmission in the presence of an optimizing bank with various interest rates. Moreover, the banking sector developed in their paper differs from that developed here in four important ways. First, the bank's deposit creation

[11] I refrain from using the term *financial sector* when describing the lender/borrower relationships in these papers since many of them omit banks or intermediaries entirely, and focus on direct finance between savers and borrowers. Moreover, most of the literature also omits money and therefore commercial banks. This is important since a crucial role of banks is the production of money—i.e., deposits—when issuing loans.

[12] It is worth mentioning, however, Pfeifer (2016) reestimates Jermann and Quadrini (2012)'s model correcting capital stock timing—among other modeling issues—and finds that shocks to the marginal efficiency of investment displace financial shocks as the dominant source of volatility among real variables.

technologies are fundamentally different. Their bank produces loans based on labor effort and collateral values whereas the bank's FMC technology in my paper depends on the bank's liquidity holdings and an exogenous supply shifter. Second, the type of bank loans offered are different. Their bank loans are for consumption expenditures, while the bank loans in this paper are working capital loans on the production side of the economy. Third, their bank does not accumulate net worth. And fourth, my paper estimates bank loan supply shocks. Jakab and Kumhof (2019), like this paper, focus on the macroeconomic effects of FMC shocks. They find that FMC shocks can have a sizeable effect on the economy. Unlike this paper, however, Jakab and Kumhof (2019) do not explicitly model the unemployment rate or various other labor market variables. Additionally, their FMC technology is black-boxed and their model is not estimated.

The macroeconomic labor SAM literature goes back to Pissarides (1987) and Mortensen and Pissarides (1994). These papers attempt to explain observed cyclical variation in labor market variables. For example, Leduc and Liu (2020) introduce time-varying search and recruiting intensities to improve the model fit, while Tsasa (2022) introduces a nonlinear production function with capital and finds that it reduces the model's ability to match labor market volatility. Taking a different approach, Petrosky-Nadeau (2014) assumed that unmatched vacant firms must obtain external finance to fund their cost of posting a vacancy. As in Bernanke et al. (1999), Petrosky-Nadeau (2014) assumes agency costs between lenders and borrowers giving rise to financial frictions in credit markets. These financial frictions allow for higher persistence in labor market variables leading to a better empirical fit than the standard SAM model of unemployment. Using a New Keynesian framework with labor SAM frictions, Gertler, Sala, and Trigari (2008) and Christiano, Eichenbaum, and Trabandt (2016) introduce a novel wage-setting condition showing that it can also improve the model's fit of business cycle dynamics.

My paper is closely related to Mumtaz and Zanetti (2016), Zanetti (2019) and Monacelli et al. (2023) who also study financial disruptions in a model with labor search frictions. Adding a labor search and matching block to the model proposed in Jermann and Quadrini (2012), Zanetti (2019) finds that financial shocks have sizeable effects on financial variables and wages, and that job destruction shocks play an important role in describing unemployment rate fluctuations. On the other hand, Monacelli et al. (2023) impose an endogenous borrowing constraint on firms and study the implications of the wage bargaining channel on employment. Like my paper, these papers also allow for disturbances within the financial sector to potentially affect real variables such as employment. However, my paper differs from theirs in an important way. My paper explicitly models the banking sector that is consistent with McLeay et al. (2014), Werner (2014) and Bundesbank (2017) descriptions of banks leading to financial frictions originating from the bank's proprietary FMC technology. In this way, the mechanism linking bank credit supply to the external finance premium is similar to that in Langlais (2023a). As a result, nominal bank deposits *and* the

loan interest rate play a critical role in the transmission mechanism between the disturbances in the banking sector and the real economy.

My paper is also related to Wasmer and Weil (2004) and Petrosky-Nadeau and Wasmer (2013). These papers impose search frictions on both labor and credit markets leading to a financial accelerator mechanism capable of propagating and/or amplifying shocks much like the standard financial accelerator models. Unlike my paper, however, the focus of these papers is on improving the fit of SAM models when financial frictions are added under the assumption banks only intermediate pre-accumulated funds.

Empirical papers that have studied the role of bank credit and labor market outcomes are vast. For example, studying the effects of credit contractions on 20 OECD countries, Borsi (2018) finds that periods of credit contractions are associated with a one-percentage-point rise in the general unemployment rate while the effect on youth unemployment rate is a rise of 2.4 percentage points. From the perspective of credit expansions, Acosta and Cortés (2022), using Mexican firm-bank level data, find that a one standard deviation in the issuance of bank loans to firms leads to a 2.6% increase in a firm's employment. Similarly, Jeong (2023) studies the role of government programs aimed at improving access to credit for small business owners in disadvantaged communities, and finds that better access to credit increases firm employment. Chodorow-Reich (2014) finds that the credit supply channel explains one-third to one-half of the employment decline in small and medium-sized firms during the Great Recession. Using bank-firm Spanish data, Alfaro, García-Santana, and Moral-Benito (2021) find that bank credit supply shocks have sizeable propagation effects, especially during the financial crisis, on employment, investment, and output.

3 Empirical Evidence

3.1 Bank Credit Supply Shock

Obtaining an exogenous bank credit supply shifter requires finding a source of variation that banks respond to when deciding to issue loans, and that this source of variation does not simultaneously influence loan demand. As a consequence, estimating a bank credit supply shock can be challenging. Fortunately, there exists a bank credit supply measure for the U.S. economy. Bassett et al. (2014) (BCDZ henceforth) derived a measure of bank credit supply from U.S. commercial bank SLOOS data. Banks that participate in the SLOOS are asked to report if they have changed their lending standards—such as collateral requirements, risk tolerance etc.—of various types of loans from the previous period. Importantly, the SLOOS asks about the direction of changes in lending standards—e.g., BCDZ group responses in one of three groups: eased, tightened, or remained unchanged—but not the magnitude of change. The result is a *qualitative* measure of bank credit supply as opposed

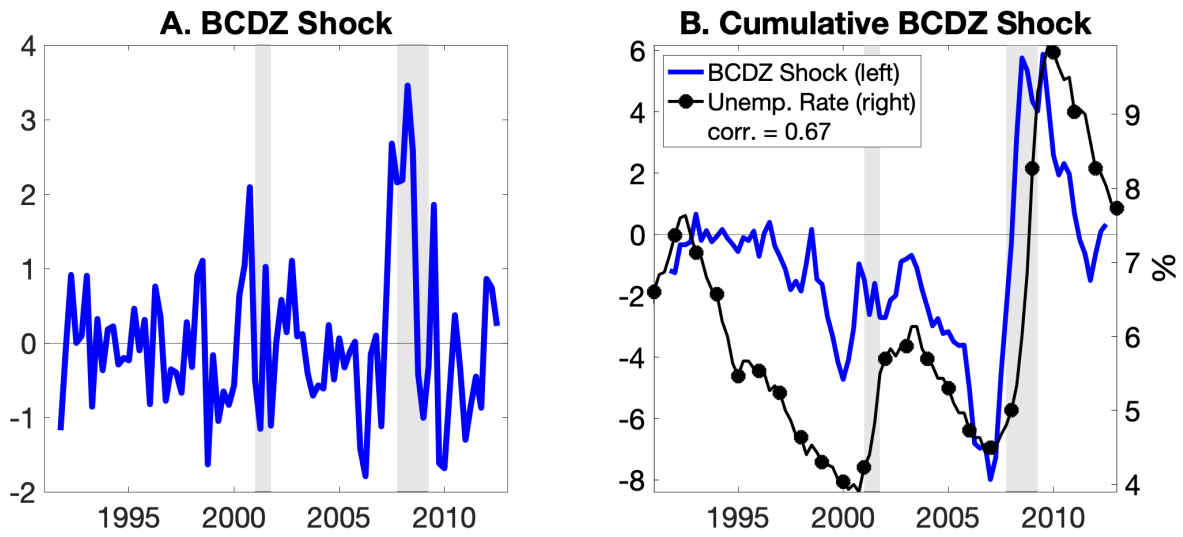

Figure 2: BCDZ Bank Credit Supply Shock

Notes: Shaded bars denote NBER recessions.
Source: Bassett et al. (2014); Bureau of Labor Statistics; author's calculation.

to a *quantitative* measure. Importantly, the SLOOS also asks banks about any change in demand for various types of loans. BCDZ construct their bank credit supply shock by purging the bank-level lending officer survey responses of macroeconomic, projection/outlook, and banking sector factors believed to influence loan demand. An asset-weighted aggregated measure of their series is publicly available on the Federal Reserve's website.[13] I refer to this bank credit supply shock series as the 'BCDZ shock'.

Figure 2 panel A plots the BCDZ shock spanning the period 1991:Q4–2012:Q3. By construction, a positive (negative) value denotes a tightening (easing) of lending standards indicative of a contraction (expansion) in bank credit supply. Generally speaking, the raw series exhibits very little autocorrelation. Notwithstanding, just prior to the 2001 and 2007-09 recessionary episodes the BCDZ series predicts three and six periods of sustained tightening of lending standards, respectively. This cyclical pattern is further revealed when the cumulative sum of the raw BCDZ shock is plotted against the unemployment rate (panel B). A strong positive relationship reveals itself suggesting exogenous shifts in the bank credit supply curve may potentially contain information regarding employment dynamics. This is particularly true surrounding the 2007-09 financial crisis. The boom period between 2003 through 2006 was characterized by easing credit conditions and a falling unemployment rate. By 2007, however, the boom quickly reversed as the unemployment rate began

[13]For further information regarding the BCDZ shock series, see https://doi.org/10.1016/j.jmoneco.2013.12.005; whereas the data files can be found here: https://www.federalreserve.gov/econres/feds/changes-in-bank-lending-standards-and-the-macroeconomy.htm. For further information regarding the SLOOS, see https://www.federalreserve.gov/data/sloos.htm.

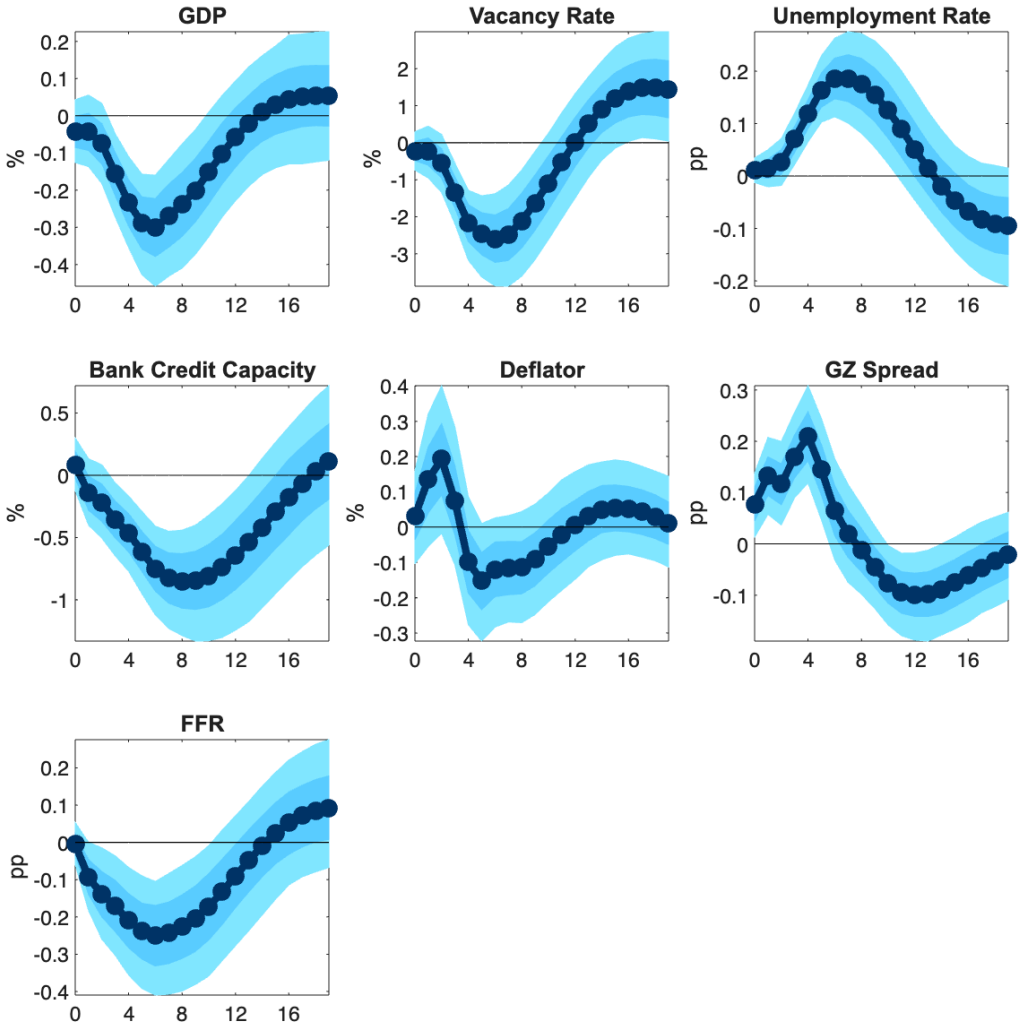

Figure 3: Impulse Responses to Contractionary BCDZ Shock

Notes: Light shaded and dark shaded areas denote 95% and 68% confidence intervals, respectively. Confidence intervals are computed using 2,000 bootstrap draws. See text for further details.

to rise sharply soon after an unprecedented tightening of lending standards.

3.2 VAR Model

In this subsection I estimate the empirical significance that bank credit supply shocks have on the unemployment rate using a monetary VAR model. The VAR model consists of seven endogenous variables and an updated version of the BCDZ shock taken from Cavallo, Morelli, Zarutskie, and Baylor (2024).[14] The endogenous variables (in order) are log real GDP, log vacancy rate,

[14]The updated BCDZ series of Cavallo et al. (2024) can be found here: https://doi.org/10.17016/2380-7172.3516

unemployment rate, log bank credit capacity, log GDP deflator, credit spread and the shadow federal funds rate taken from Wu and Xia (2016). Bank credit capacity is the sum of total bank loans and unused credit commitments. The measure of credit spread used is the 'GZ' spread.[15] I include the 'GZ' spread to capture potential financial shocks originating outside of the banking system while the federal funds rate controls for the stance of monetary policy. As discussed in Bassett et al. (2014) and Cavallo et al. (2024), since the BCDZ shock has been purged of macroeconomic and loan demand side factors, the endogenous variables should not have any influence on the BCDZ shock through loan demand. As a result, I order the BCDZ shock first in the VAR model. Under this identification scheme the bank credit supply shock can potentially influence the remaining seven variables within one quarter. The VAR model is estimated using standard OLS techniques with three lags and a quadratic time trend.[16] Sample data is quarterly and spans the period 1992:Q1–2019:Q4.

Table 1: VAR Variance Decompositions

horizon	GDP	Vacancy Rate	Unemp. Rate	Bank Credit Capacity	Deflator	GZ Spread	Shadow Federal Funds Rate
$h = 4$	12	13	12	11	8	30	13
	[2–30]	[2–30]	[2–29]	[2–28]	[1–26]	[13–48]	[1–32]
$h = 8$	37	42	47	35	13	34	29
	[14–58]	[18–63]	[25–66]	[13–58]	[3–28]	[18–52]	[7–53]
$h = 16$	32	36	38	35	14	32	29
	[12–54]	[16–56]	[17–60]	[12–60]	[4–29]	[16–50]	[9–51]
$h = 32$	27	33	32	26	16	30	27
	[10–48]	[14–53]	[14–55]	[8–51]	[5–31]	[14–47]	[8–49]

Notes: Each cell entry denotes the percentage contribution attributed to the bank credit supply shock in explaining the volatility of the column variable at horizon h. Entries in square brackets denote the 95% confidence interval based on 2,000 bootstrap draws.

Figure 3 displays the impulse responses of the endogenous variables due to a one standard deviation increase in the updated BCDZ shock. Following the bank credit supply shock—corresponding to an unexpected contraction in the supply of bank credit—the economy experiences a contraction consistent with the recession phase of a typical business cycle. Notwithstanding, the economy responds very little on impact. Only after about two quarters following the shock does the economy begin contracting. By the two-year horizon, bank credit capacity contracts by 0.85% and the unemployment rate rises by about 0.2 percentage points. Moreover, the cost of borrowing (GZ spread) rises and firms post fewer vacancies. The delayed response to the tightening of bank lending standards has been documented in the previous literature. For example, Gilchrist and Zakrajšek (2012a) find that precommitted lines of credit, which are held off-balance sheet, play an important role in credit markets over the business cycle. When banks begin tightening lending standards, credit

[15] See Table 8 of the Appendix for a complete description of the data and sources.
[16] Lag length was selected based on AIC criteria.

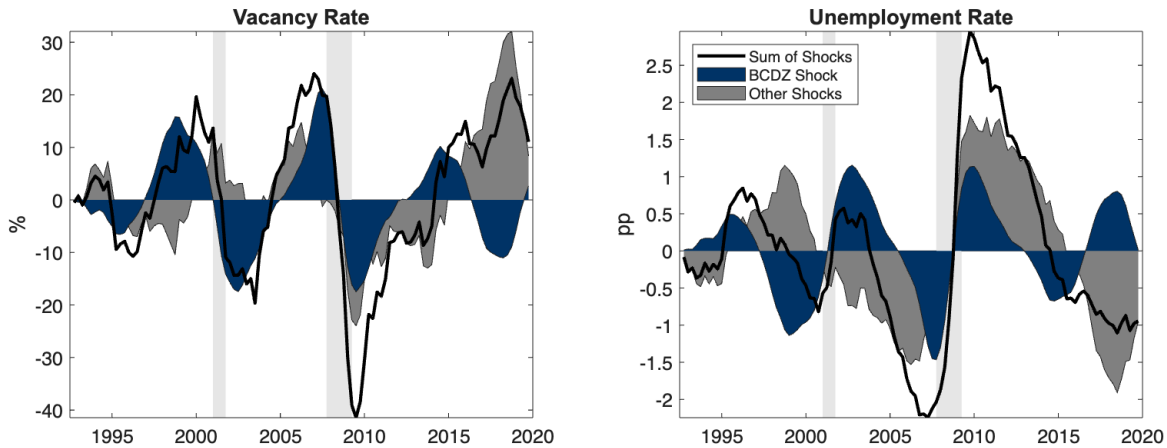

Figure 4: Historical Decomposition of the Vacancy and Unemployment Rate

Notes: Shaded bars denote NBER recessions. Decomposition based on VAR model estimates. See text for further details.

constrained firms and households begin to draw down credit lines which may lead to a temporary stagnation of spending and bank credit capacity before beginning to contract.[17]

While Figure 3 suggests bank credit supply shocks can affect the unemployment rate in the predicted direction, it does not quantify the bank credit supply shock's relative importance in generating the observed volatility of the unemployment rate. Table 1 shows the variance decomposition performed on the same VAR model. Each cell entry denotes the percentage contribution of the BCDZ shock at generating the variation of the endogenous variables. Numbers in square brackets denote 95 percent confidence intervals. Consistent with the impulse response results, at short horizons (e.g., four quarters) the BCDZ shock plays a limited role in generating economic volatility. However, around the two-year horizon, the BCDZ shock accounts for 42 and 47 percent of the volatility in the vacancy rate and unemployment rate, respectively. Except for the GDP deflator, the BCDZ shock can explain at least 29% of the volatility of the remaining endogenous variables at the same horizon.

How have bank credit supply shocks affected key labor market variables historically? Consider the historical decomposition of firm vacancy rate and the unemployment rate shown in Figure 4. The solid black line represents the deviation of the series from its trend, while the blue shaded areas denote the deviations of each series due to the BCDZ shock only. In general, the BCDZ shock plays an important role in generating the observed cyclical pattern in the data. This is especially true during the years surrounding the 2001 and 2007 recessionary periods.

[17]This is apparent when comparing bank credit growth (blue line) in Figure 1 panel A and the SLOOS (blue line) in Figure 1 panel B. Bank credit growth begins to contract only after banks begin tightening lending standards.

Robustness Test: To test whether the BCDZ shock accurately captures exogenous variation in the bank credit supply curve, I estimate an alternative bank credit supply shock using a Bayesian VAR model with sign restrictions. The details of the estimation procedure are available in the online Appendix section C.1. After performing identical analysis with the alternative bank credit supply measure, I find that the results are remarkably similar to those presented above—especially during the period around the financial crisis.

Overall, these results suggest that bank credit supply shocks are an important determinant of the unemployment rate and, more broadly, a driver of economic volatility in the recent past. Next, I build a model to rationalize these empirical results.

4 Model

In this section I derive a discrete time infinite horizon DSGE model economy which rationalizes the empirical findings outlined above: that bank credit supply shocks play an important role in describing the observed comovement between outstanding bank credit and the unemployment rate. Three critical ingredients in the model are (*i*) labor market search frictions, (*ii*) nominal price and wage rigidities, and (*iii*) that the banking sector is consistent with the FMC view. Since the money supply and bank credit in the model are treated as nominal variables, these assumptions allow bank credit supply shocks to potentially contribute to the short-term volatility of the unemployment rate over the business cycle.

The model closely follows Leduc and Liu (2016). Different than Leduc and Liu (2016), however, this paper includes capital accumulation, money, and a banking sector. There are five main sectors of the economy: household, production, banking, government, and a monetary authority. The household sector consists of a continuum of infinitely lived and identical households with unit measure. The representative household consists of a continuum of worker members who, at any moment, are either employed or unemployed. The household accumulates the currency and capital stock, and owns each firm and the representative bank. The production sector consists of three types of firms: a continuum of intermediate-good firms, a continuum of monopolistically competitive retail firms, and a representative final-goods firm. Each intermediate-good firm uses one worker and some capital to produce a homogeneous intermediate good. Each retail sector firm then uses intermediate goods to produce a differentiated output good which it sells to the final-goods firm. The final-goods firm combines all the differentiated goods into a final good that can be consumed, invested in the capital stock, used up for capital utilization, used up when posting a vacancy or used as bank fees.

In each period, the share of workers who are unemployed must search for jobs with the intermediate-good firms that have posted a vacancy. Moreover, each time an intermediate-good

firms post a vacancy they incur a fixed cost. Successful matches are produced with a matching technology that transforms searching workers and vacancies into an employment relation. Following a successful match, the intermediate-good firm then rents some capital from the household. Wages are determined by Nash bargaining between a searching worker and a hiring firm, while the capital rental rate is determined in a competitive capital market. In each period all matched firms must obtain a bank loan to pay their labor wage and capital rental bills in advance of production.

4.1 Bank & Money Supply

The banking sector in this paper is a modified version of that developed in Langlais (2023a) and is composed of a representative bank that is risk neutral and accumulates net worth. At the beginning of period t the bank has net worth ($NetWorth_t$) in money units. The bank offers two types of deposit services to agents in the economy—demand deposits and savings deposits—and extends loans to intermediate-good firms. The bank is situated in a fractional reserve system and is licensed by the monetary authority to operate a proprietary *financing through money creation* (FMC) technology. The FMC license allows the bank's short-term liabilities—i.e., demand deposits—to trade on par with the government-issued currency. In other words, the FMC technology gives the bank the ability to create part of the economy's money supply through the bank's issuance of new demand deposits. As a result, the economy's agents view demand deposits as money and use it—along with currency in circulation—for transaction purposes and settlement of any outstanding debt. The savings deposits, however, are non-transaction deposits. Thus, the supply of money is comprised of currency in circulation, M_t, and total bank demand deposits.

At the beginning of each period the household deposits a share of their currency holdings with the bank. In return for the currency, the bank issues the household savings deposit, D_t^s. Of which the bank is required by the monetary authority to hold fraction $\rho^s \in [0,1]$ of these savings deposits as reserves.[18] As a result, total currency in circulation just before the period t production cycle is

$$M_t = M_t^{\text{base}} - D_t^s, \tag{1}$$

where M_t^{base} is the total currency base held by the household entering period t. Towards the end of the production cycle, however, the monetary authority injects or removes $X_t \in [-M_t^{\text{base}}, \infty)$ units of currency from the economy. As a result, towards the end of the period total currency in circulation is $M_t + X_t$, and total currency entering period $t+1$ is $M_{t+1}^{\text{base}} = M_t^{\text{base}} + X_t$.

Each intermediate-good firm i who has successfully matched with a worker is required to pay

[18] Since I do not explicitly model the monetary authority's balance sheet, the bank does not have deposit—i.e., reserve accounts—with the monetary authority. As a consequence, all *reserves* consist of money base held by the bank—i.e., currency not in circulation.

their labor wage and capital rental bills in advance of production.[19] As a result, each operating firm must obtain a loan from the bank that is financed by two different sources of funds: newly created demand deposits and bank net worth. Since the bank does not directly lend out currency from its reserves—and there is no incentive for agents to withdraw their demand deposits as currency—part of the demand deposits are backed by the household's currency deposits. As a result, the portion of firm i's bank loan financed by demand deposits is denoted D_{it}^d and in aggregate is determined by the bank's FMC technology. Namely, total demand deposits used to finance loans to matched firms are

$$\int_{i\in\mathscr{M}_t} D_{it}^d \mathrm{d}i = \underbrace{\zeta_{bt} \times E_t}_{\text{FMC}} + \underbrace{E_t}_{\text{backed by currency deposits}}, \tag{2}$$

where ζ_{bt} denotes the FMC shock which follows a stationary AR(1) process and E_t denotes the bank's excess reserves holdings. The term $\mathscr{M}_t$ denotes the set of matched firms during the period t production cycle. Moreover, like the savings deposits, the monetary authority requires the bank to hold a minimum amount of reserves equal to fraction $\rho^d \in [0,1]$ total demand deposits. Unlike savings deposits, however, the FMC technology allows the bank to choose its desired demand deposit reserve ratio, denoted $\tilde{\rho}_t^d$, so long as $\tilde{\rho}_t^d \geq \rho^d$. The term ζ_{bt} in equation 2 represents an exogenous bank credit supply shifter meant to capture shifts driven by factors not related to liquidity—i.e., excess reserve—position.

The bank's balance sheet immediately following the issuing of loans, and before the end of the period t production cycle, is shown in Table 2.

Table 2: Bank Balance Sheet Prior to Production

Assets	Liabilities
loans: $\int_{i\in\mathscr{M}} D_{it}^d \mathrm{d}i + NetWorth_t$	deposits: $D_t^s + \int_{i\in\mathscr{M}} D_{it}^d \mathrm{d}i$
reserves: $\rho^s D_t^s + \tilde{\rho}_t^d \int_{i\in\mathscr{M}} D_{it}^d \mathrm{d}i$	$NetWorth_t$

Using the balance sheet identity, it is straightforward to show that total reserves are equal to total savings deposits, D_t^s, and given the bank can choose its desired demand deposit reserve ratio—insofar as it chooses a ratio above the minimum required—it holds an amount of excess reserves, $E_t \geq 0$. Formally,

$$\underbrace{D_t^s}_{\text{total reserves}} = \underbrace{\rho^s D_t^s + \rho^d \int_{i\in\mathscr{M}_t} D_{it}^d \mathrm{d}i}_{\text{required reserves}} + \underbrace{E_t}_{\text{excess reserves}},$$

[19] Unlike Christiano, Motto, and Rostagno (2010), I assume, that each firm must finance 100% of their variable costs with a bank loan instead of 75%—in the U.S. case—or 92%—in the EU case—proposed by Christiano et al. (2010).

where excess reserves are found to be

$$E_t = (1-\rho^s)D_t^s - \rho^d \int_{i\in\mathscr{M}_t} D_{it}^d \mathrm{d}i, \tag{3}$$

where the first term on the right-hand-side denotes unencumbered reserves associated with savings deposits and the second term denotes the (minus) total required reserves of the deposits created by the bank's FMC technology.

Although the bank chooses how much excess reserves to hold each period, due to the functional form assumptions of the bank's FMC technology 2, the bank's desired reserve ratio collapses to an expression solely dependent on the exogenous FMC shock. Proposition 4.1 shows that the bank's time-varying desired reserve ratio is inversely related to the FMC shock.

Proposition 4.1 *The bank's desired demand deposit reserve ratio, $\tilde{\rho}_t^d$, can be written as*

$$\tilde{\rho}_t^d = \frac{1}{1+\zeta_{bt}} + \rho^d. \tag{4}$$

This result shows the link within the banking sector between FMC and its desired amount of liquidity holdings. As banks curtail lending, fewer deposits are created to finance loans implying a higher desired reserve ratio. Not only does this mechanism determine the quantity of deposits available to finance loans to firms, as shown below, this mechanism is crucial for understanding the dynamics of credit spreads—i.e., the cost of external finance. The bank's *money multiplier* is shown in Corollary 4.1.1.

Corollary 4.1.1 *The bank's money multiplier—i.e., the ratio of total demand deposits to savings deposits—is found as*

$$\frac{1-\rho^s}{\tilde{\rho}_t^d}. \tag{5}$$

I leave the proofs in Appendix A.

Following the period t production cycle settlement of all contracts occurs. The bank's balance sheet a moment after the production cycle, but just before the end of the period, is displayed in Table 3.

Table 3: Bank Balance Sheet Following Production

Assets	Liabilities
loans: $(1+R_t^l)\left[\int_{i\in\mathscr{M}} D_{it}^d \mathrm{d}i + NetWorth_t\right]$	deposits: $(1+R_t^s)D_t^s + (1+R_t^d)\int_{i\in\mathscr{M}} D_{it}^d \mathrm{d}i$
reserves: $\rho^s D_t^s + \tilde{\rho}_t^d \int_{i\in\mathscr{M}} D_{it}^d \mathrm{d}i$	$NetWorth_t + \Pi_t^b$

Interest rates R_t^l, R_t^s, and R_t^d denote the net nominal interest rate on loans, savings deposits and demand deposits, respectively. The variable Π_t^b denotes the bank's net profits and is found as the value of end-of-period assets minus the value of end-of-period liabilities and less beginning-of-period net worth:

$$\Pi_t^b = (1+R_t^l)\left[\int_{i\in\mathscr{M}_t} D_{it}^d \mathrm{d}i + NetWorth_t\right] + \rho^s D_t^s + \tilde{\rho}_t^d \int_{i\in\mathscr{M}_t} D_{it}^d \mathrm{d}i$$
$$-(1+R_t^s)D_t^s - (1+R_t^d)\int_{i\in\mathscr{M}_t} D_{it}^d \mathrm{d}i - NetWorth_t. \quad (6)$$

Optimal behavior of the bank requires choosing $\{\{D_{it}^d\}_{i\in\mathscr{M}_t}, E_t, D_t^s\}$ to maximize profits (6) subject to the FMC technology (2) and excess reserves (3). When the bank is behaving optimally, the credit spread—denoted $R_t^l - R_t^d$—is shown in proposition 4.2

Proposition 4.2 *The bank's optimality conditions yield the interest rate spread between the loan rate, R_t^l, and demand deposit rate, R_t^d:*

$$R_t^l - R_t^d = \frac{\tilde{\rho}_t^d}{1-\rho^s} R_t^s. \quad (7)$$

I leave its proof in Appendix A. Proposition 4.2 shows the crucial pricing link between the bank's desired reserve ratio, $\tilde{\rho}_t^d$, and the firms' cost of borrowing, R_t^l. As discussed below, this is a key transmission mechanism in the model linking the dynamics between the (nominal) banking sector and the firms' demand for capital and labor, and therefore, the economy's unemployment rate. Since the demand deposit rate is governed by the monetary authority and the savings deposit rate governs the household's intertemporal consumption-savings choice (see below), when the bank contracts loans and increases its desired reserve ratio, the main margin of adjustment is the loan interest rate.

After the production cycle and towards the end of the period, the bank pays a share $\tau \in (0,1)$ of gross profits as a dividend payment to the household. After which, the bank then receives lump-sum banking fees, $P_t\phi^b$, from the household. Here P_t denotes the money price of the final good and $\phi^b > 0$ denotes an amount of final goods. As a result, the bank's net worth evolves according to

$$NetWorth_{t+1} = (1-\tau)\left\{\Pi_t^b + NetWorth_t\right\} + P_t\phi^b,$$

where the first term denotes the remaining share of gross profits—apart from fees—retained by the bank.

4.2 Households

The representative household owns and accumulates the economy's aggregate raw capital stock, K_t, and aggregate currency, M_t^{base}. After observing period t state, the household chooses the capital utilization rate, $\varkappa_t$, which turns K_t units of raw capital into $\varkappa_t K_t$ units of capital services. Capital services are rented to intermediate-good firms in a competitive market. The household receives utility from consumption of the final output good, C_t, but receives disutility proportional to the share of members who work, n_t. As a result, share $1-n_t$ of members are unemployed. To motivate public holdings of money, additional utility is gained by holding real currency value, M_t/P_t, during period t production cycle. The representative household's preferences are as follows:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ \log(C_t - hC_{t-1}) - \psi_L n_t + \overbrace{\mathscr{V}\left(\frac{M_t}{P_t}\right)}^{\text{real currency holding preference}} \right\}, \tag{8}$$

where $\beta \in (0,1)$ denotes the household's subjective discount factor, $h \in (0,1)$ measures consumption habit persistence, and $\psi_L > 0$ denotes the disutility weight on employment. The function $\mathscr{V}(\cdot)$ is increasing in its argument and governs the preference of liquidity holdings of real currency.

The household receives income from labor services, $W_t n_t$, and capital rental services, $P_t r_t^k \varkappa_t K_t$, where W_t and r_t^k denote the nominal wage rate and real capital rental rate, respectively. Additionally, the household receives unemployment benefit income, $P_t \mathscr{B}(1-n_t)$, deposit income on previous period's savings deposits, $(1+R_{t-1}^s)D_{t-1}^s$, as well as currency injection from the monetary authority, X_t. The term $\mathscr{B} > 0$ denotes an amount of final goods.

Following Christiano et al. (2010), I assume the currency injection is received by the household after period t deposit decisions are made. Different than the literature, however, I assume that income generated by savings deposits are only available for use in the following period.[20] The household uses income and currency holdings to purchase consumption goods, $P_t C_t$, investment goods, $P_t I_t$, pay for capital utilization cost $P_t a(\varkappa_t)$, and to accumulate next period's currency, M_{t+1}^{base}. As a result, the household's asset accumulation equation looks like,

$$\begin{aligned} M_{t+1}^{\text{base}} \leq\ & W_t n_t + [\varkappa_t r_t^k - a(\varkappa_t)] P_t K_t + P_t \mathscr{B}(1-n_t) \\ & + (1+R_{t-1}^s) D_{t-1}^s + M_t + X_t + \text{Lump}_t - P_t C_t - P_t I_t, \end{aligned} \tag{9}$$

where Lump_t denotes lump-sum transfers which is composed of bank and firm profits, as well as taxes/subsidies and bank fees. The function $a(\varkappa_t)$ captures the convex capital utilization cost in

[20] This assumption further distinguishes *savings* deposits from *demand* deposits and, importantly, is a technical assumption needed to ensure a model solution given the values of calibrated parameters discussed below. Thus, the bank's profits 6 are measured using accrual accounting.

terms of final goods. The aggregate capital stock evolves according to

$$K_{t+1} = (1-\Delta)K_t + \left[1 - S(\frac{I_t}{I_{t-1}})\right] I_t, \tag{10}$$

where $\Delta \in (0,1)$ denotes the depreciation rate of capital and $S(\cdot)$ is a convex function representing capital installation costs in real terms.

Each period the household chooses a bundle $\{C_t, M_{t+1}^{\text{base}}, D_t^s, K_{t+1}, I_t, M_t, \varkappa_t\}$ that maximizes lifetime utility (8) subject to the currency constraint (1), asset accumulation (9) and law of motion of capital (10). The household's formal optimization problem is in Appendix C.2.2.

4.3 Aggregation Sector

At each time t a composite final good, Y_t, is produced by a perfectly competitive representative firm. The firm produces the composite good by combining a continuum of retail goods, indexed by $j \in (0,1)$, using technology

$$Y_t = \left[\int_0^1 Y_{jt}^{\frac{1}{\lambda_f}} \mathrm{d}j\right]^{\lambda_f}, \quad 1 \leq \lambda_f < \infty, \tag{11}$$

where Y_{jt} denotes the time t output of retail-good firm j. The parameter λ_f governs the degree of price mark-up over marginal cost for retail-good firms. As a result of profit maximization, the demand for retail good j is

$$Y_{jt} = Y_t \left(\frac{P_{jt}}{P_t}\right)^{\frac{\lambda_f}{1-\lambda_f}}, \tag{12}$$

where P_{jt} is the per unit money price of retail good j. Integrating 12 and imposing 11 yields a relationship between the final good's price and retail good prices:

$$P_t = \left[\int_0^1 P_{jt}^{\frac{1}{1-\lambda_f}} \mathrm{d}j\right]^{1-\lambda_f}. \tag{13}$$

4.4 Retail-Good Firms

There is a continuum of retail-good firms who each make differentiated retail good Y_{jt} with linear production technology

$$Y_{jt} = \chi_{jt}, \tag{14}$$

where χ_{jt} is the input of intermediate good used by retail firm j. While the retail goods firms are price takers in the input market, they are monopolistic competitors in the product markets. As a

result, retail goods firms set prices for their products, taking as given the demand schedule 12 and the price index 13.

As in Rotemberg (1982), I assume that each firm j's price is subject to quadratic adjustment costs and is proportional to nominal GDP:

$$\frac{\xi_p}{2}\left(\frac{P_{jt}}{P_{jt-1}}-\pi\right)^2 P_t Y_t,$$

where the parameter $\xi_p \geq 0$ governs the cost of adjusting prices in period t and π denotes the steady-state (gross) inflation rate, where $\pi_t = P_t/P_{t-1}$. As a result, retail firm j's nominal profits can be written as

$$\begin{aligned}\Pi^f_{jt} &= P_{jt}Y_{jt} - Q_t\chi_{jt} - \frac{\xi_p}{2}\left(\frac{P_{jt}}{P_{jt-1}}-\pi\right)^2 P_t Y_t,\\ &= \left(P_{jt}-Q_t\right)\underbrace{Y_t\left(\frac{P_{jt}}{P_t}\right)^{\frac{\lambda_f}{1-\lambda_f}}}_{=\chi_{jt}} - \frac{\xi_p}{2}\left(\frac{P_{jt}}{P_{jt-1}}-\pi\right)^2 P_t Y_t, \end{aligned} \tag{15}$$

where Q_t denotes the competitive price of the intermediate good. To arrive at equation 15 I substitute out χ_{jt} using 14 and impose retail firm j's output demand 12. Thus, in each period t retail firm j chooses P_{jt} to maximize expected real lifetime profits:

$$\mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \frac{\lambda_{1,t+s}}{\lambda_{1,t}} \frac{\Pi^f_{jt+s}}{P_{t+s}},$$

subject to 15. The term $\beta^s \frac{\lambda_{1,t+s}}{\lambda_{1,t}}$ denotes the household's stochastic discount factor where $\lambda_{1,t} = P_t\tilde{\lambda}_{1,t}$ with $\tilde{\lambda}_{1,t}$ being the Lagrange multiplier on the household's asset accumulation equation 9. In a symmetric equilibrium, $P_{jt} = P_t\ \forall j$, and the optimality condition is

$$q_t = \frac{1}{\lambda_f} + \xi_p \frac{\lambda_f - 1}{\lambda_f}\left[\pi_t(\pi_t-\pi) - \beta\mathbb{E}_t\left\{\frac{\lambda_{1,t+1}}{\lambda_{1,t}}\frac{Y_{t+1}}{Y_t}\pi_{t+1}(\pi_{t+1}-\pi)\right\}\right] \tag{16}$$

where $q_t = Q_t/P_t$ and denotes the real marginal cost. Under no price adjustment cost, i.e., $\xi_p = 0$, the real marginal cost is simply the inverse of the markup.

4.5 Labor Market

At the beginning of each period t, there are u_t unemployed workers searching for jobs and there are v_t vacancies posted by intermediate-good firms. Following Mortensen and Pissarides (1994), the

number of successful worker-firm matches occur according to a constant returns to scale matching function $\mu(v_t, u_t)$. The probability of an open vacancy matching with a searching worker is

$$p^v(\vartheta_t) \equiv \frac{\mu(v_t, u_t)}{v_t} = \mu(1, 1/\vartheta_t),$$

and the probability of an unemployed and searching worker matching with a firm posting an open vacancy is denoted

$$p^u(\vartheta_t) \equiv \frac{\mu(v_t, u_t)}{u_t} = \mu(\vartheta_t, 1), \tag{17}$$

where $\vartheta_t = \frac{v_t}{u_t}$ denotes labor market tightness.

At the beginning of period t there are n_{t-1} workers. Of these workers, fraction δ separate from their job leaving $(1-\delta)n_{t-1}$ workers remaining employed. Following job separation—but before the production cycle—last period's unemployed workers and newly separated workers begin to search for jobs with match probability 17. Consequently, the total number of job searchers in period t evolves according to

$$u_t = 1 - n_{t-1} + \delta n_{t-1}. \tag{18}$$

As in Blanchard and Galí (2010) and Leduc and Liu (2016), I assume newly matched workers begin working in the period the match is made. Moreover, I assume full participation and define the unemployment rate as the fraction of the population who are left without a job after hiring takes place in period t. As a result, aggregate employment in period t evolves according to

$$n_t = (1-\delta)n_{t-1} + \mu(v_t, u_t). \tag{19}$$

while the unemployment rate, $\mathscr{U}_t$, is found as

$$\mathscr{U}_t = u_t - \mu(v_t, u_t) = 1 - n_t. \tag{20}$$

4.6 Intermediate-Good Firm

There is a continuum of identical intermediate-good firms indexed by i with a common production technology $F(k_{it}, n_{it}; z_t)$ that is homogeneous of degree one, increasing in both capital services, k_{it}, and labor input, n_{it}, and is concave, and continuously differentiable. The term z_t is a common TFP term that follows a log-AR(1) process. Each firm can only produce output if she has successfully hired a worker. Moreover, in each period a hired worker supplies the firm with exactly one unit of

labor. As a result, firm i's output following period t's production cycle is

$$\chi_{it} = \begin{cases} z_t f(k_{it}), & \text{if matched} \\ 0, & \text{otherwise.} \end{cases}$$

where $z_t f(k_{it}) \equiv F(k_{it}, 1; z_t)$ and $f' > 0$ and $f'' < 0$. Since the wage bill and capital rental costs must be paid in advance of production, a successfully matched intermediate-good firm must obtain a loan from the bank to finance its operational costs. Thus, total loans given to firm i are equal to $W_t + P_t r_t^k k_{it}$.

A matched firm obtains a flow profit of $Q_t z_t f(k_{it}) - (1 + R_t^l)[W_t + P_t r_t^k k_{it}]$ in the current period. In following period, the match survives with probability $1 - \delta$ and the firm continues producing; with probability δ the match is destroyed and the firm may post a new job vacancy. As a result, the Bellman equation for a matched firm is written as

$$J_{it}^F = \max_{k_{it}} Q_t z_t f(k_{it}) - (1 + R_t^l)[W_t + P_t r_t^k k_{it}] + \beta \mathbb{E}_t \left\{ \frac{\tilde{\lambda}_{1,t+1}}{\tilde{\lambda}_{1,t}} \left[(1-\delta) J_{it+1}^F + \delta J_{it+1}^V \right] \right\}, \tag{21}$$

where J_{it}^F is the value of a matched firm and J_{it}^V is the value of an unmatched firm who posted a vacancy. Following a successful match, a firm then chooses its quantity of capital services to employ that satisfies $Q_t z_t f'(k_{jt}) = (1 + R_t^l) P_t r_t^k$. Since firms are price takers in the capital service rental market, I find that $k_{it} = k_{jt}\ \forall\ i \neq j$, which implies $J_{it}^F = J_{jt}^F$ and $J_{it}^V = J_{jt}^V\ \forall\ i \neq j$. As a result, I drop intermediate-good firm index subscripts. Appendix C.2.3 derives the firms' optimization problem for a representative intermediate-good firm.

A firm who is unmatched and posts a new vacancy in period t must pay nominal cost $P_t \kappa$, where $\kappa > 0$ is a quantity of final goods. With probability $p^v(\vartheta_t)$ the firm obtains a match and the vacancy is filled yielding value J_t^F. Otherwise, with probability $1 - p^v(\vartheta_t)$ the vacancy remains into the following period yielding the value J_{t+1}^V. Thus, the Bellman equation for a firm with a vacancy is written as

$$J_t^V = -P_t \kappa + p^v(\vartheta_t) J_t^F + \beta \mathbb{E}_t \left\{ \frac{\tilde{\lambda}_{1,t+1}}{\tilde{\lambda}_{1,t}} (1 - p^v(\vartheta_t)) J_{t+1}^V \right\}. \tag{22}$$

4.7 Worker Value Functions

During period t, an employed worker receives wages W_t in monetary units but sustains a utility cost—in monetary terms—of $P_t \psi_L / \lambda_{1,t}$. In the following period, if the match is sustained the worker receives value J_{t+1}^W; whereas if the match is destroyed the worker begins their search for a new match. With probability $p^u(\vartheta_{t+1})$ a worker searching for a job obtains a new match and

receives value J^W_{t+1}, while a worker who is unsuccessful at obtaining a match receives value J^U_{t+1} for being unemployed. As a result, the Bellman equation for an employed worker is written as

$$J^W_t = W_t - P_t \frac{\psi_L}{\lambda_{1,t}} + \beta \mathbb{E}_t \left\{ \frac{\tilde{\lambda}_{1,t+1}}{\tilde{\lambda}_{1,t}} \left[(1-\delta) J^W_{t+1} + \delta \left\{ p^u(\vartheta_{t+1}) J^W_{t+1} + (1 - p^u(\vartheta_{t+1})) J^U_{t+1} \right\} \right] \right\}. \tag{23}$$

An unemployed worker obtains flow benefits $P_t \mathscr{B}$ from the government and can search for a job in period $t+1$. With probability $p^u(\vartheta_{t+1})$ an unemployed worker successfully matches with a firm and becomes employed and with probability $1 - p^u(\vartheta_{t+1})$ she remains unemployed and receives value J^U_{t+1}. As a result, the Bellman equation for an unemployed worker is written as

$$J^U_t = P_t \mathscr{B} + \beta \mathbb{E}_t \left\{ \frac{\tilde{\lambda}_{1,t+1}}{\tilde{\lambda}_{1,t}} \left[p^u(\vartheta_{t+1}) J^W_{t+1} + (1 - p^u(\vartheta_{t+1})) J^U_{t+1} \right] \right\}. \tag{24}$$

4.8 Nash Bargaining & Wage Rigidity

Workers and intermediate-good firms bargain over wages by solving the following problem

$$\max_{W_t} \left\{ \left(J^W_t - J^U_t \right)^b \left(J^F_t \right)^{1-b} \right\}, \tag{25}$$

where $b \in (0,1)$ denotes the workers' bargaining power. The first-order condition looks like

$$b J^F_t \frac{\partial \left(J^W_t - J^U_t \right)}{\partial W_t} + (1-b) \left(J^W_t - J^U_t \right) \frac{\partial J^F_t}{\partial W_t} = 0.$$

The solution to the bargaining problem is shown in proposition 4.3.

Proposition 4.3 *The solution to the Nash bargaining problem gives the worker surplus, $J^W_t - J^U_t$, and firm surplus, J^F_t, as*

$$J^W_t - J^U_t = \omega(R^l_t; b) S_t \quad \text{and} \quad J^F_t = \Big(1 - \omega(R^l_t; b) \Big) S_t,$$

respectively, where

$$\omega(R^l_t; b) \equiv \frac{b}{b + (1-b)(1+R^l_t)},$$

and $S_t \equiv J^F_t + J^W_t - J^U_t$ denotes the total net surplus value generated by a match in period t.

I leave its proof in the Appendix A. Unlike the standard SAM literature without working capital

loans, the equilibrium surplus shares depend on the loan interest rate R_t^l. It is clear to see that when the firms' cost of borrowing rises, the worker's share of surplus falls. In this sense the credit market frictions allow part of the firms' borrowing costs to be transferred to workers. Notwithstanding, the economic effect of this wedge is insignificant.[21]

Subtracting 24 from 23 and using proposition 4.3, the Nash bargaining wage is determined by the following relationship

$$\omega(R_t^l;b)S_t = W_t^N - P_t\frac{\psi_L}{\lambda_{1,t}} - P_t\mathscr{B} + \beta\mathbb{E}_t\left\{\frac{\lambda_{1,t+1}}{\lambda_{1,t}}(1-\delta)\left(1-p^u(\vartheta_{t+1})\right)\omega(R_{t+1}^l;b)S_{t+1}\right\}, \quad (26)$$

where W_t^N denotes the wage which solves the Nash bargaining problem 25.

As in Leduc and Liu (2016), I assume a wage rigidity between the equilibrium market wage and the Nash bargaining wage. Specifically, I assume

$$\frac{W_t}{P_t} = \left(\frac{W_{t-1}}{P_{t-1}}\right)^{\iota}\left(\frac{W_t^N}{P_t}\right)^{1-\iota},$$

where $\iota \in [0,1]$ is a model parameter which governs the mix between the previous period's market wage and the Nash bargaining wages.

4.9 Policy, Market Clearing & Search Equilibrium

Each period the government's fiscal policy is limited to collecting lump-sum taxes, Tax_t, from the household which it uses to exclusively finance the unemployment benefits of unemployed workers. As a result, the period t government budget constraint is written as

$$P_t\mathscr{B}(1-n_t) = Tax_t.$$

The monetary authority conducts monetary policy according to two different rules. The first is a money base growth rule that takes the following form:

$$x_t - x = \rho_x(x_{t-1} - x) + \varepsilon_t^x$$

where $x_t = \frac{X_t}{M_t^{\text{base}}}$ and x denotes the steady-state money base growth rate and ρ_x denotes the persistence of money base growth rate. The term ε_t^x represents an unanticipated monetary policy shock to the money base growth rate that follows an exogenous process described below. The second rule is an interest rate rule governing the short-term demand deposit rate that takes the form of a general

[21] Assuming a bargaining weight of $b = 0.5$, under the benchmark calibration with steady-state interest rate $R^l = 0.011$ implies a bargaining share of $\omega = 0.497$, whereas in the standard model $\omega = 0.5$.

Taylor Rule:

$$R_t^d - R^d = \rho_r(R_{t-1}^d - R^d) + (1-\rho_r)\left[\alpha_\pi^r \log(\frac{\mathbb{E}_t \pi_{t+1}}{\pi}) + \alpha_y \log(\frac{Y_t}{Y_{t-1}})\right] + \varepsilon_t^r,$$

where R^d denotes the steady-state deposit interest rate and ρ_r governs the rate's persistence. The parameters α_π and α_y denote the policy weights on the deviation of expected inflation from its steady-state rate and GDP growth rate respectively. The term ε_t^r represents an unanticipated monetary policy shock to the deposit rate that follows an exogenous process described below.

In a search equilibrium, the markets for final consumption goods, intermediate goods, capital services, and loans all clear. The final good's market clearing condition yields the resource constraint

$$C_t + I_t + \kappa v_t + \frac{\xi_p}{2}(\pi_t - \pi)^2 Y_t + a(\varkappa_t)K_t + \phi^b = Y_t.$$

Capital services market clearing implies $\int_{i\in\mathscr{M}_t} k_{it}\mathrm{d}i = \varkappa_t K_t$, where the intermediate-good's market clearing condition implies $Y_t = \int_{i\in\mathscr{M}_t} z_t f(k_{it})\mathrm{d}i$. Finally, the bank loan market clearing condition implies

$$NetWorth_t + \int_{i\in\mathscr{M}_t} D_{it}^d \mathrm{d}i = \int_{i\in\mathscr{M}_t} \left\{W_t + P_t r_t^k k_{it}\right\}\mathrm{d}i. \tag{27}$$

With free entry of intermediate-good firms, in equilibrium $J_t^V = 0$ for all t and 22 implies that

$$\frac{\kappa}{p^v(\vartheta_t)} = \frac{J_t^F}{P_t}. \tag{28}$$

4.10 Calibration, Steady-State & Comparative Statics

I assume the household's real liquidity preference has the following functional form

$$\mathscr{V}(x) = \begin{cases} \gamma\frac{(x)^{1-\sigma_q}}{1-\sigma_q}, & \text{if } \sigma_q \neq 1 \\ \gamma\log(x) & \text{if } \sigma_q = 1, \end{cases}$$

where $\gamma > 0$ and σ_q are parameters which govern the utility weight and utility curvature of real money, respectively. Following Christiano et al. (2010) I assume capital utilization cost function takes the following form

$$a(\varkappa_t) = \frac{r^k}{\sigma_a}\left\{\exp[\sigma_a(\varkappa_t - 1)] - 1\right\},$$

Table 4: Calibrated Parameters

Variable	Description	Value	Source/Target
α_f	production function curvature	1/3	standard
Δ	capital depreciation rate	0.025	standard
α	matching function elasticity	0.5	Leduc and Liu (2016)
δ	job destruction rate	0.1	Leduc and Liu (2016)
b	Nash bargaining weight	0.5	Leduc and Liu (2016)
$\mathscr{B}$	unemployment benefits	0.25	Leduc and Liu (2016)
ρ^s	saving deposit reserve ratio	0.0	Feinman (1993)
ρ^d	demand deposit reserve ratio	0.1	Feinman (1993)
τ	bank dividend rate	0.05	Langlais (2023a)

Notes: One period corresponds to one quarter.

where r^k denotes the steady-state capital rental rate in real terms and $\sigma_a > 0$ is a parameter governing the cost function convexity. As is standard in the search literature, I assume the number of successful worker-firm matches, $\mu(v_t, u_t)$, is governed by a Cobb-Douglas matching technology:

$$\mu(v_t, u_t) = \bar{\mu} u_t^{\alpha} v_t^{1-\alpha},$$

where the parameters $\bar{\mu} > 0$ governs the matching efficiency and $\alpha \in (0,1)$ governs the elasticity of job matches with respect to the number of searching workers. Following Christiano et al. (2014) I assume the investment adjustment cost function, $S(\cdot)$, takes the following functional form

$$S(x) = \frac{1}{2}\left\{\exp\left[\sqrt{S''}(x-1)\right] + \exp\left[-\sqrt{S''}(x-1)\right] - 2\right\},$$

and has the following properties: $S(1) = S'(1) = 0$, $S''(1) = S'' > 0$, where S'' is a model parameters governing the curvature of the function. The intermediate-good firms' production function takes the standard Cobb-Douglas form

$$F(k_t, n_t; z_t) = z_t k_t^{\alpha_f} n_t^{1-\alpha_f}, \quad \text{implying} \quad z_t f(k_t) = z_t k_t^{\alpha_f}, \quad \alpha_f \in (0,1).$$

The model's parameters are grouped into three categories: calibrated, targeted and estimated. For reference, one period in the model corresponds to one quarter. The calibrated parameters are set to values which are common in the related literature and are listed in Table 4. As is standard, production function curvature on capital, α_f, and capital depreciation rate, Δ, are set to 1/3 and 0.025 (10% per year), respectively. Following Leduc and Liu (2016), I set the elasticity of matching, α, to 0.5, job separation rate, δ, to 0.1, Nash bargaining weight, b, to 0.5, and unemployment benefits, $\mathscr{B}$, to 0.25. Following the Federal Reserve's minimum reserve requirements for non-transaction and transaction deposits, I set the minimum reserve ratios for savings deposits, ρ^s, and demand deposits,

Table 5: Targeted Steady-State Variables & Endogenous Parameters

Variable	Description	Value	Source/Target
Panel A: Targeted Steady-State Variables			
U	unemployment rate	6.4%	Leduc and Liu (2016)
$\mathscr{B}+\frac{\psi_L}{\lambda_1}$	total benefit of non-work	0.75	Leduc and Liu (2016)
π	inflation rate	1.005	2% (apr)
$\frac{K}{Y}$	capital-output ratio	10.7	Christiano et al. (2010)
p^v	vacancy filling rate	0.7	den Haan, Ramey, and Watson (2000)
$\frac{\kappa}{GDP}$	vacancy cost share	0.02	2% of GDP
D^s/M^{base}	savings deposit share	0.09	currency to money base ratio = 91%[a]
R^s	savings deposit rate	5% annual	3% real rate
R^l-R^d	interest rate spread	1.84% annual	GZ spread[b]
$R^d<R^l<R^s$	interest rate term structure	—	upward sloping yield curve
bank leverage	assets to net worth ratio	13.6	financial sector leverage[c]
Panel B: Endogenous Parameters			
β	discount rate	0.998	
x	steady-state money base growth	0.005	
ζ_b	steady-state FMC	2.731	
$\bar{\mu}$	matching function efficiency	0.644	
κ	vacancy cost	0.017	
λ_f	mark-up	1.175	
ψ_L	labour disutility	0.532	
z	steady-state productivity	0.564	
γ	liquidity preference weight	0.031	
σ_q	liquidity preference curvature	1.200	
ϕ^b	lump-sum banking fees	0.005	

Notes: Unless stated otherwise, sample averages are computed over the sample period 1987:Q1-2008:Q2.
[a] Currency component of M1 divided by monetary base. Source: Board of Governors of the Federal Reserve System H.6 and H.3, FRED codes CURRSL and MBCURRCIR.
[b] Sample average of the 'GZ spread'. Source: Gilchrist and Zakrajšek (2012b).
[c] Sample average of financial sector assets to equity. Financial sector consists of chartered depository institutions, security brokers and dealers, finance companies, credit unions, and life insurance companies. Source: Board of Governors of the Federal Reserve System under Z1/OTHER.

ρ^d, to be 0% and 10%, respectively.[22] Finally, I set the bank dividend rate, τ, to 5%.

Table 5 lists the second group of parameters (panel B) which are endogenously determined based on targeted steady-state moments (panel A) in the data or literature. Data for computing sample averages span the period 1987:Q1-2008:Q2. Given a targeted annual nominal savings rate of 5% and annual inflation rate of 2%, the household's subjective discount factor, β, is found to be 0.998. The steady-state inflation rate of 2% annually implies a steady-state growth rate of the money base, x, to be 0.005. I set the steady-state level of the FMC shock, ζ_b, such that the interest rate spread, R^l-R^d, matches the sample average 'GZ' spread in the data. Following Leduc and Liu (2016) I set the steady-state unemployment rate to 6.4% and the steady-state vacancy filling

[22] Although the Federal Reserve has eliminated all minimum reserve requirements as of 2020, over the period in which the model is estimated minimum reserve requirements on transaction deposits were positive. https://www.newyorkfed.org/medialibrary/media/research/epr/02v08n1/0205bennpdf.pdf

rate, p^v, to 0.7. This implies that the matching function efficiency parameter, $\bar{\mu}$, is 0.644. The cost of posting a vacancy, κ, is set to 2% of steady-state GDP. As a result, I find $\kappa = 0.017$. The firms' mark-up parameter, λ_f, is set such that the steady-state term structure is upward sloping: $R^d < R^l < R^s$. This results in a mark-up of 17.5% over marginal cost. Following Leduc and Liu (2016) I set the steady-state utility benefit of non-work to 0.75, implying a labor disutility weight, ψ_L, of 0.532. The steady-state level of TFP, z, is set such that the steady-state capital-output ratio, K/Y, is 10.7 as in Christiano et al. (2010). However, the steady-state level of TFP also depends on the consumption habit persistence parameter, h. As a consequence, z can only be attained once h has been estimated.[23] A targeted steady-state savings deposit to money base ratio, D^s/M^{base}, of 0.09 and interest rate, R^s, yield liquidity preference weight parameter, γ, and liquidity curvature parameter, σ_q, of 0.031 and 1.20, respectively. The steady-state ratio D^s/M^{base} is computed as $1 - \frac{M}{M^{\text{base}}}$, where $\frac{M}{M^{\text{base}}}$ is the sample average of currency in circulation to base money ratio. Finally, bank leverage is targeted to 13.6 implying real banking fees, ϕ^b, of 0.005;

The remaining parameters—all of which do not affect the model's steady state—are estimated using Bayesian estimation techniques in Section 5.1.[24] For reference, I refer to the model with parameter value set to those in Tables 4, 5 and 6 (see below) as the benchmark model.

Comparative Statics Before I estimate the remaining parameters, I pause to describe the key relationship of this paper in a static context: that between bank credit supply and the equilibrium unemployment rate in the model's non-stochastic steady state. To show this I perform two exercises. First, I formally derive a steady-state condition which must be met for the relationship between the level of bank FMC and the unemployment rate to be consistent with the empirical observation. Second, I numerically show that the model's steady state does satisfy this condition thereby leading to a inverse relationship between the level of FMC and the unemployment rate. Furthermore, I show how various model parameters affect this credit-unemployment nexus.

The main result of the first exercise can be summarized in proposition 4.4.

Proposition 4.4 *If the steady-state loan rate of interest, R^l, is sufficiently responsive to the level of FMC, ζ_b, then the steady-state unemployment rate, $\mathscr{U}$, is decreasing in the level of FMC. Specifically, if $\frac{\partial R^l}{\partial \zeta_b} < \Xi \frac{\partial \lambda_1}{\partial \zeta_b} \Rightarrow \frac{\partial \mathscr{U}}{\partial \zeta_b} < 0$, where*

$$\Xi \equiv \frac{(1+R^l)\frac{\psi_L}{\lambda_1^2}}{\left(\frac{K}{n}\right) r^k + \frac{\psi_L}{\lambda_1} + \mathscr{B}},$$

[23] Once h is known, z is found using a solution algorithm described in the Online Appendix C.4.

[24] There is, however, one exception: the consumption habit persistence parameter, h, which affects the steady-state level of money's marginal utility.

and variables without "t" subscript denote steady-state values and λ_1 denotes the Lagrange multiplier on the household's asset accumulation equation 9.

I leave its formal proof in Appendix A and provide a descriptive analysis here.

Consider this relationship graphically. Panel A of Figure 5 plots the job-creation curve and wage curve under two different levels of bank FMC: $\zeta_b = 2.731$ (solid line) and $\zeta_b = 2.731/10$ (dashed line). Below, panel B plots the associated Beveridge curve. When ζ_b is reduced, R^l rises shifting both curves downward (dashed lines). However, the effect on the job-creation curve is relatively stronger leading to a fall in the labor market tightness.[25] As in the standard labor-search models, a lower market tightness is associated with a higher unemployment rate through the Beveridge curve.

Next, I plot the steady-state relationship between the level of FMC and the unemployment rate under several different calibrations (see Figure 6). Specifically, I change the values of required demand deposit reserve ratio (panel A), steady-state inflation rate (panel B), matching efficiency (panel C), and unemployment benefits (panel D). As expected, under the benchmark calibration (solid lines) the relationship between the level of FMC and the unemployment rate is negative.

When the monetary authority raises the minimum required reserve ratio on demand deposits (dotted line panel A), the unemployment rate rises for each level of FMC. This result highlights the fact that bank credit supply is also governed by forces outside the bank's control—in this case by policy. Increasing ρ^d while holding ζ_b constant reduces the bank's excess reserves and increases the loan interest rate (see Propositions 4.1 and 4.2). This reduces total loans available to firms to purchase their inputs and increases their borrowing costs. Both effects lead to a reduction in demand for labor and capital services for matched firms. As a result, labor market tightness falls and the unemployment rate rises.

When the steady-state inflation rate is raised from 2% to 3%, holding ζ_b constant, this increases the steady-state unemployment rate. This is because the inflation rate determines the nominal savings rate faced by the household—i.e., R^s—and this rate is a determinant of the interest rate spread (see Proposition 4.2).[26] A higher interest rate spread is associated with higher cost of borrowing for the firms. As a result, labor market tightness falls and the unemployment rate rises.

Consistent with the standard labor-search models without financial frictions, a rise in matching efficiency or a rise in unemployment benefits have the predicted effect on the steady-state unemployment rate. A higher matching efficiency, for a given level of ζ_b, increases the probability of workers and firms finding a match. As a result, more matches are made and the unemployment rate falls. On the other hand, when unemployment benefits rise, again holding ζ_b constant, the value of

[25] For expositional purposes, the downward shift in the Nash Wage curve (dashed line) is accentuated.

[26] Combining the household's first-order conditions with respect to D_t^s and M_{t+1}^{base}, yields steady-state savings rate of $R^s = \left(\frac{\pi}{\beta}\right)^2 - 1$. See Online Appendix C.4 for further details.

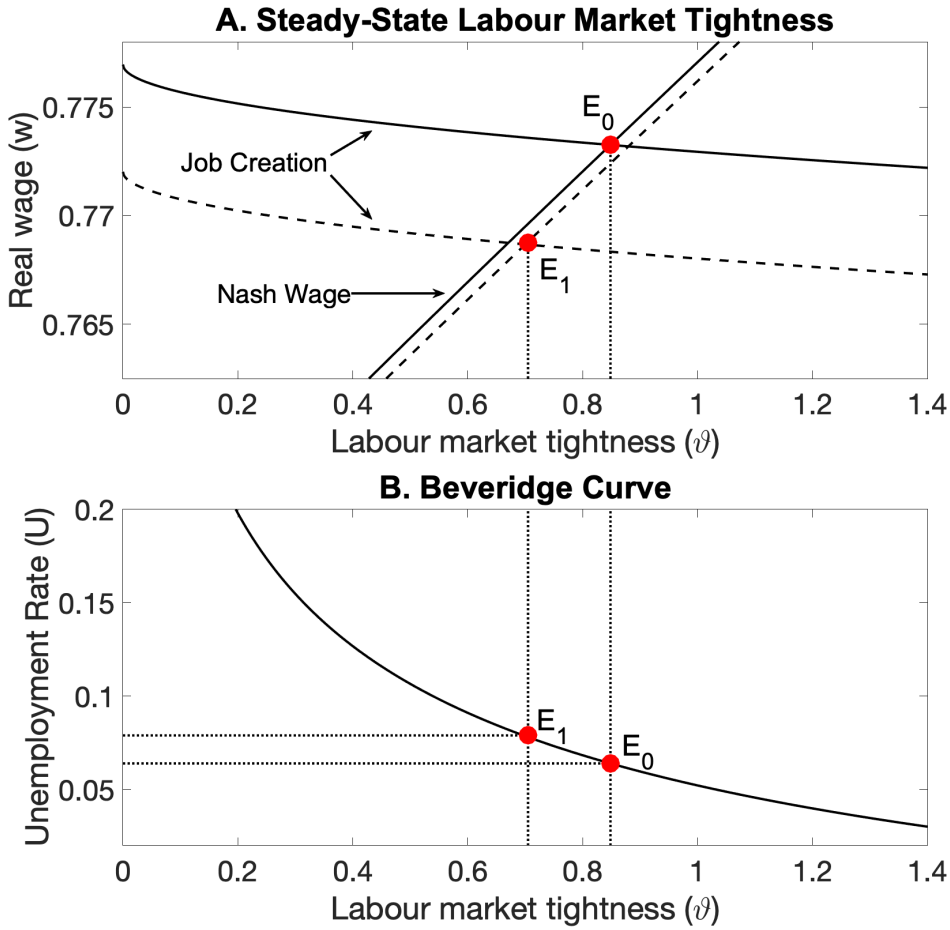

Figure 5: Comparative Statics—Reduction in the level of FMC (ζ_b)

Notes: Comparative statics are computed for parameter values listed in Tables 4 and 5. I assume $h = 0.641$. Equilibrium E_0 is associated with $\zeta_b = 2.731$, whereas equilibrium E_1 is associated with $\zeta_b = 2.731/10$. I assume a constant capital-labour ratio and constant marginal utility of money under both equilibria. The downward shift in the Nash Wage curve (dashed line) is accentuated for expositional purposes.

unemployment rises which reduces the surplus generated from a match. As a result, labor market tightness falls and unemployment rate rises.

5 Dynamic Analysis

In this section I describe the macroeconomic data used to estimate the remaining parameters of the model and analyze the model's dynamic properties in response to bank credit supply shocks. The primary objective of this section is to quantify the relative importance that bank credit supply shocks have on the unemployment rate.

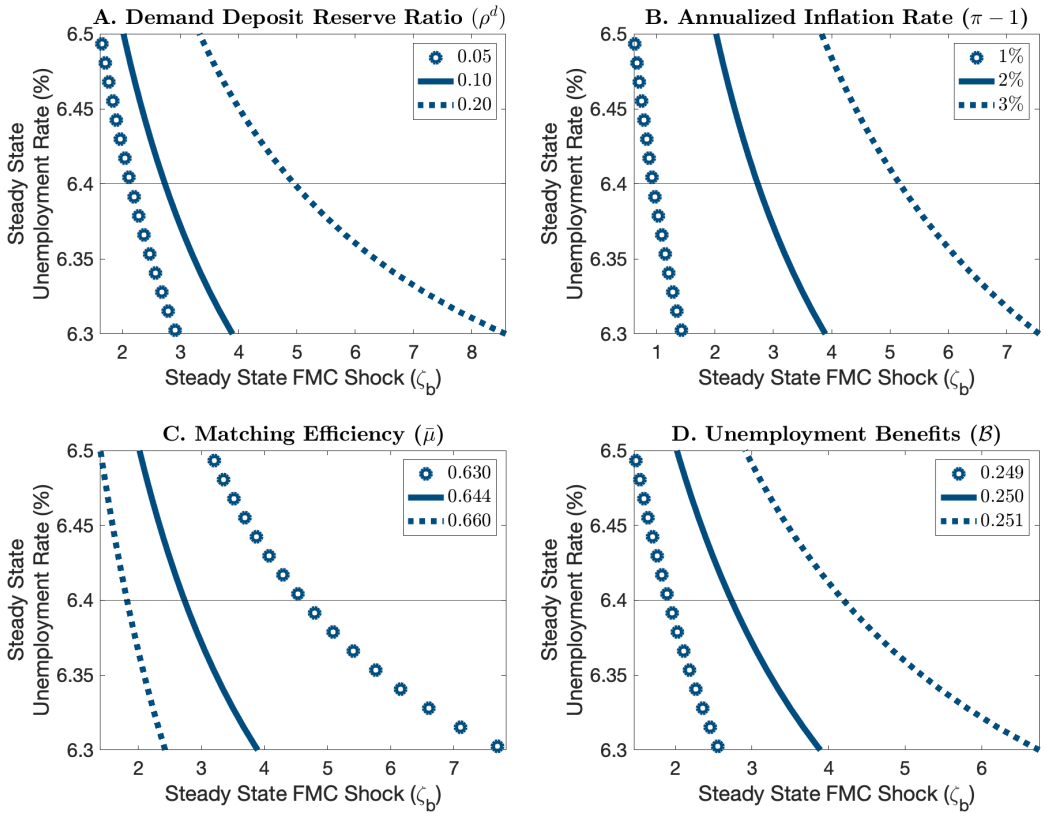

Figure 6: Comparative Statics

Notes: Comparative statics are computed for parameter values listed in Tables 4 and 5. I assume $h = 0.641$.

5.1 Data & Parameter Estimation

I use Bayesian techniques to obtain estimates of the mode and posterior distribution of each estimated parameter. Except for the household's consumption habit persistence parameter, h, the estimated parameter does not affect the model's steady state. The dataset is comprised of six macroeconomic variables, in quarterly frequency, describing the U.S. economy over the period 1989:Q1–2019:Q4.[27] The variables are: real per capita GDP growth, firm vacancy rate, unemployment rate, inflation rate, external finance premium and the federal funds rate. The real per capita GDP is logged then first-differenced. The vacancy rate is logged and demeaned. Inflation rate is computed as the first difference of the logged GDP implicit price deflator. The measure used for the external finance premium is the 'GZ' spread. All series then have their respective sample means removed to ensure consistency with their model counterpart. Further details regarding dataset construction and sources

[27]The actual starting period is 1985:Q1 and was chosen to minimize the impact of potential structural breaks from the 'Volker Period'. The first 16 quarters are utilized as a training sample for the Kalman filter iterations. As a result, the likelihood estimation is performed over the period 1989:Q1–2019:Q4

see Section B of the Online Appendix.

I assume there are four structural shocks in the model: the two monetary policy shocks already discussed, ε_t^x and ε_t^r, as well as ε_t^b and ε_t^z where

$$\log(\frac{\zeta_{b,t}}{\zeta_b}) = \rho_b \log(\frac{\zeta_{b,t-1}}{\zeta_b}) + \varepsilon_t^b,$$
$$\log(\frac{z_t}{z}) = \rho_z \log(\frac{z_{t-1}}{z}) + \varepsilon_t^z,$$

where $\varepsilon_t^j \sim \mathcal{N}(0, \sigma_j^2)$ and $\sigma_j > 0$ for $j \in \{r, x, b, z\}$. Parameters $\{\sigma_j\}$ and $\{\rho_j\}$ are estimated. In addition to the four shocks, I include a measurement errors on the observed external finance premium. This is motivated by the fact that the model, and therefore the model counterpart to the external finance premium—i.e., the interest rate spread $R^l - R^d$—does not incorporate any risk or information about economic uncertainty which the 'GZ' spread has been shown to contain (Gilchrist and Zakrajšek, 2012b). As a result of not having risk or uncertainty in the model's loan market, the measurement error placed on the 'GZ' spread can be thought of as a misspecification factor of the model in failing to capture observed economic uncertainty in the loan market. To avoid stochastic singularity, I also place a measurement error on the observed vacancy rate. Notwithstanding, the estimation procedure will still be useful in quantifying the relative importance that the FMC shocks have on the observed external finance premium and vacancy rate.

Table 6 lists the estimated parameters, their associated prior distributions and their posterior distribution's mode and standard deviations. Figures 16 and 17 of the Online Appendix plot the parameter priors and posterior distributions. For each analyses performed below I set each estimated parameter to its respective posterior mode.[28]

5.2 Model Response to Bank Credit Supply Shock

What happens following a contraction in the supply of bank credit? The solid blue line in Figure 7 represents the model response to an unexpected one-standard deviation contraction in the bank's FMC. A deep recession ensues following the shock: GDP and bank loans contract by about 1.25 and 1.5 percent, respectively, while the unemployment rate rises 1.1 percentage points at the one year mark. Since the estimated FMC shock process is persistent, the economic recession persists beyond the five-year horizon. Notwithstanding, consumption returns to its steady-state level around the four-year mark. Relative to the monetary VAR model results, the FMC shock has a stronger affect on the economy. This could be a result of the parsimonious nature of the model which omits potentially other important aggregate demand shocks such as uncertainty shocks (Christiano et al.,

[28] Initial estimation results indicate $\hat{\sigma}_a \approx 0$. As a consequence, the estimation results presented here I calibrate σ_a to 0.0001.

Table 6: Estimated Parameters—Baseline Model

		Prior			Posterior	
		Type	Mean	Std. dev.	Mode	Std. dev. (Hess.)
Panel A: Economic Parameters						
h	Habit persistence	Beta	0.4	0.1	0.583	0.038
ξ_p	Price adj. cost	Norm.	80	50	49.298	6.173
S''	Investment adj. cost curvature	Norm.	5	3	15.442	2.012
σ_a	Capital utilization cost curvature	Gamma	6	5	-	-
ι	Real wage rigidity	Beta	0.5	0.15	0.956	0.004
Panel B: Monetary Policy Parameters						
ρ_r	Interest rate smoothing	Beta	0.75	0.1	0.944	0.010
α_π	Interest rate weight on inflation	Norm.	1.5	0.25	1.667	0.217
$\alpha_{\Delta y}$	Interest rate weight on output growth	Norm.	0.25	0.1	0.249	0.090
ρ_x	Money base smoothing	Beta	0.5	0.1	0.567	0.105
Panel C: Shock Autocorrelations						
ρ_z	Firm technology	Beta	0.5	0.2	0.802	0.013
ρ_b	FMC technology	Beta	0.5	0.2	0.974	0.006
Panel D: Shock Standard Deviations & Measurement Error						
σ_z	Technology	Invg.	0.01	0.05	0.0087	0.001
σ_b	Inter-period FMC technology	Invg.	0.03	0.3	1.1043	0.125
σ_r	Monetary policy rate	Invg.	0.58	0.825	0.2783	0.019
σ_x	Money base growth injection	Invg.	0.01	0.05	0.0077	0.002
	Std. dev. measurement error on spread	Weibull	0.01	5	0.0049	0.0004
	Std. dev. measurement error on vacancy rate	Weibull	0.01	5	0.1427	0.0092

Notes: Invg. denotes 'Inverse gamma type 1'. Data used for estimating parameters represents the US economy and span the period 1985:Q1–2019:Q4 (first 16 quarters are used for filter initialization). Data description and sources are found in Table 8 of the Appendix. The Laplace approximation of marginal density is 2567.01. The posterior distribution is obtained from 8,000,000 draws equally distributed across 8 chains of the Metropolis-Hastings algorithm. For each chain the first 20% of the draws are discarded. The parameter σ_a is set to 0.0001 prior to estimation.

2014; Leduc and Liu, 2016).

Considering the comparative static exercise performed above, the transmission mechanism generating this response is straightforward. When the bank unexpectedly contracts the supply of loans, firms are faced with a shortage of funds—i.e., money—required to purchase their inputs. As a result, demand for factor inputs falls leading to a response consistent with a typical recession: GDP, investment and vacancy rate fall, while the external finance premium and the unemployment rate rise. Bayesian IRFs for each of the shocks are shown in Online Appendix C.5.

Price & Quantity Effect There are two channels through which the FMC shock influences the model's dynamics. First, through the loan clearing condition (equation 27) via the FMC deposit creation technology (equation 2); and, second, through the bank's optimal interest rate spread (equation 7). I call the former channel the *quantity* channel and the latter the *price* channel. The red dotted line in Figure 7 shows the *price* effect by holding the $\{\zeta_{b,t}\}$ that affects the loan clearing

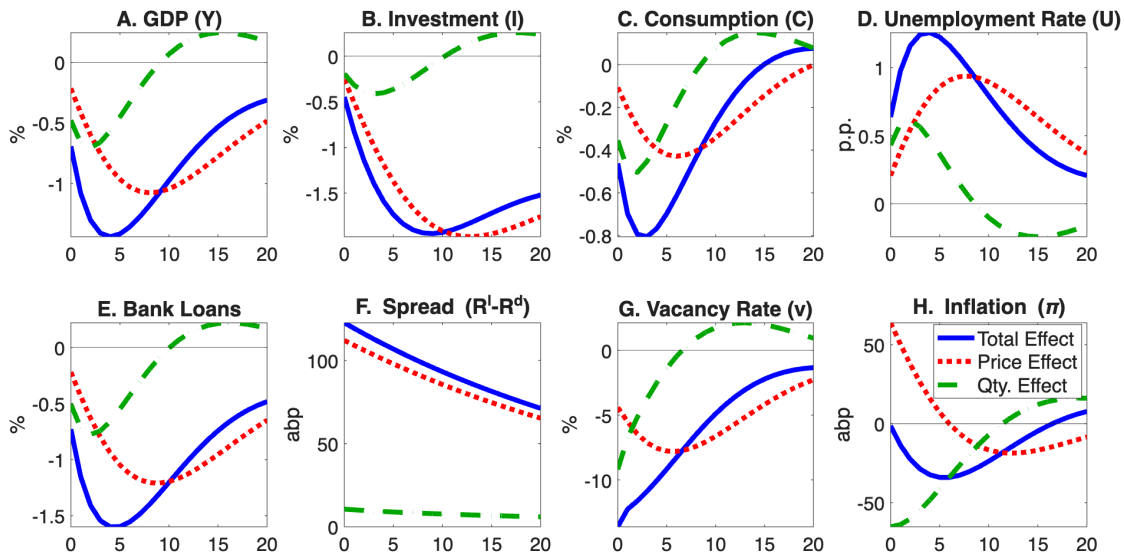

Figure 7: Impulse Responses to a FMC shock

Notes: Estimated parameters are set to their posterior mode. 'abp' denotes *annual basis points* and 'pp' denotes *percentage points*.

condition constant at its steady-state level, while allowing the $\{\zeta_{b,t}\}$ affecting the bank's optimal interest rate spread to follow its AR(1) structure following the shock. The result is a rise in borrowing costs imposed on firms without any (direct) loss of funding. Similar to the aggregate response (blue line), the rise in the interest spread lowers the firms' value of matching with a worker leading firms to post fewer vacancies and the labor market tightness—$\vartheta = v/u$—falls and the unemployment rate rises. On the other hand, the *quantity* effect (green dashed line) is estimated as the difference between the total effect and price effect.[29,30] Except for inflation rate, the *price* and *quantity* effects reinforce each other on impact. In the case of inflation, rising borrowing costs via the *price* channel pushes up firms' marginal costs leading to higher inflation, while reducing total money supply through the *quantity* channel lowers inflation. Both effects leads to overall inflation responding very little.

5.3 Variance & Historical Decomposition

Now that the mechanism linking the bank's supply of credit and the firm's hiring decisions has been established, I now compare the relative importance of each of the four structural shocks in generating the observed economic volatility. Table 7 shows the variance decomposition of each observed variables used in the Bayesian estimation (panel A) as well as other interesting variables (panel

[29]While the quantity effect operates directly through total loans available to firms, there are general equilibrium effects operating through the household's savings rate, R^s, which feedback into the interest rate spread.

[30]Since the model is solved via log linearization, the total effect is the sum of the quantity and price effects.

Table 7: Model Variance Decompositions

Variable	FMC ζ_b	TFP z	Policy Rate ε^r	Money Base Growth Rate x	ME
Panel A: observed variables					
GDP (Y)	42	22	0	35	–
Unemployment Rate ($\mathscr{U}$)	41	26	1	32	–
Policy Rate (R^d)	10	14	25	51	–
Inflation Rate (π)	5	43	1	51	–
Spread ($R^l - R^d$)	34	0	0	0	66
	[100]	[0]	[0]	[0]	[–]
Vacancy Rate (v)	31	24	0	24	21
	[39]	[30]	[1]	[31]	[–]
Panel B: other variables					
Real Credit ($wn + r^k \varkappa K$)	34	37	0	29	–
Investment (I)	52	28	1	19	–
Consumption (C)	40	18	0	41	–
Labour Market Tightness (ϑ)	41	27	1	32	–

Notes: Each cell entry denotes the percentage contribution (rounded to nearest whole number) attributed to the column shock in explaining the volatility of the row variable at a business cycle frequency (i.e., 6-32 quarters). Entries in square brackets denote variance decomposition when measurement error (ME) is omitted. Estimated parameters are set to their posterior mode.

B). Each table cell entry denotes the percentage contribution of the column shock at generating the volatility of the row variable at business cycle frequencies.[31] Surprisingly, the bank's FMC shock is the most important source of volatility for GDP (42%), unemployment (41%), vacancy rate (39% without measurement error), and interest rate spread (100% without measurement error).[32] Moreover, the FMC shock plays an important role for non-observed variables too. For example, FMC accounts for the majority of investment volatility and is the main contributor of labor market tightness volatility. Compared to the empirical estimates in Table 1, the model results suggest the estimated FMC shock plays a similar role in driving economic volatility.

Figure 8 plots the historical decomposition of the growth in the vacancy rate (panel A) and deviations in the unemployment rate from its mean (panel B) over the sample horizon. The solid black lines denote the observed data as represented by the model's response to all the estimated shocks and initial conditions. Consistent with the results above, the FMC shock (blue bars) plays an

[31] Business cycle frequencies range between 6 and 32 quarters.

[32] As mentioned above, the interest rate spread here does not incorporate economic risk or uncertainty and that the firms' net worth position is not a state variable of the optimal loan contract. As a result, looking at the variance decomposition of the spread when its measurement error is included with the FMC shock would be a more realistic estimate. In that case the variance decomposition falls to 34%.

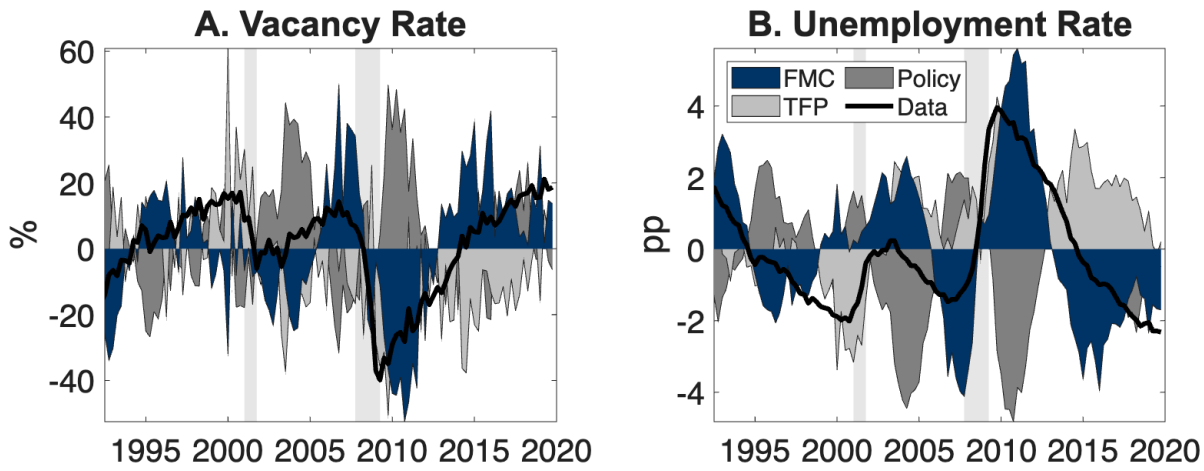

Figure 8: Historical Decomposition

Notes: Each series is demeaned. Model simulations are conducted when the estimated parameters are set to their posterior mode. Shaded areas denote NBER recession dates.

important role in generating the observed cyclicality of both the vacancy and the unemployment rates. This is particularly true during the period surrounding the 2007-09 financial crisis. Between 2005 and 2007, there were a series of positive FMC shocks driving unemployment down and vacancies up. By the time the recession began, however, the FMC shocks suddenly reversed themselves (see Figure 9 below) leading to a contraction in bank credit supply and vacancies.

The estimated monetary policy shocks—R^d and x—are countercyclical and, according to the model, played an important role at stabilizing the economy during the 2007-09 recession. Had there not been any monetary policy shocks during the financial crisis—especially shocks in the money base—the unemployment rate would have been about 4 percentage points higher, and vacancies would have experienced a longer contraction. This result should be interpreted with caution as previous papers have found that the Fed's unconventional policies following the crisis have had relatively modest effects on reducing the unemployment rate (Engen, Laubach, and Reifschneider, 2015; Chung, Gagnon, Nakata, Paustian, Schlusche, Trevino, Vilán, and Zheng, 2019; Jannsen, Potjagailo, and Wolters, 2019).

6 Out-of-Sample Tests & Alternative Identification Method

Out-of-Sample Tests: Since the FMC shock is inferred from the macroeconomic data used in the estimation procedure outlined in Section 5.1 and, therefore, is not directly observed in the data, it is possible the estimated FMC shock may be capturing other sources of variation in the economy. In this section I test to see whether the model implied FMC shock accurately captures bank credit supply shocks observed in the real world. To accomplish this task, I compare the cyclical properties

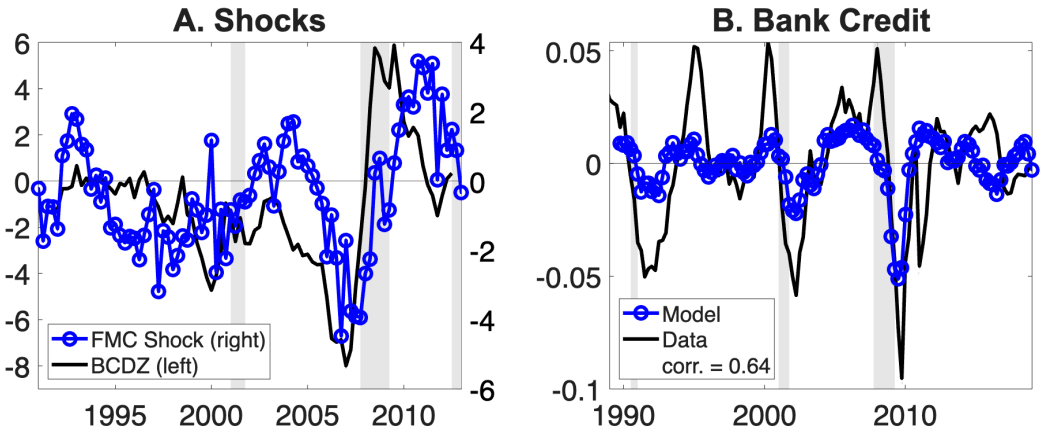

Figure 9: Out of Sample Tests

Notes: Estimated model parameters are set to their posterior mode. Shaded bars denote NBER recessions.
Sources: Bassett et al. (2014), FDIC call reports and author's calculations.

of the model implied FMC shock with the BCDZ bank credit supply shock used in Section 3. Because the BCDZ shock is estimated using *qualitative* senior loan officer opinion survey data that is not used in the model's estimation procedure makes it an ideal candidate for an *out-of-sample* test. In other words, the BCDZ series should, in theory, be measuring the same thing as the model implied FMC shock—i.e., exogenous shifts in the supply of bank credit—and, therefore, should exhibit a similar pattern over the business cycle.

Figure 9 panel A plots the cumulative sum of BCDZ shock (solid black line) and the model implied smoothed FMC shock (blue circled line). To maintain consistency between the empirical and model derived bank credit supply shocks, I plot the negative of the FMC shock implying that a rise (fall) corresponds to a contraction (expansion) in the supply of bank credit. Early in the sample period there appears to be little consistency between the shocks. From the 2004 up until the financial crisis, however, both series predict a marked increase in the supply of bank credit followed by a sudden reversal during the recessionary period.

It is important to note that data on bank credit was not included in the set of observed variables in the model estimation procedure. This means that the cyclical properties of the model implied FMC shock are largely dependent on the underlying mechanisms of the model. One could ask: *How does the estimated model's implied credit series compare to its observed counterpart*? Panel B of Figure 9 displays the model implied bank credit (blue circled line) and its observed counterpart.[33] While the model predicts a much lower level of volatility, the cyclical pattern of each series are remarkably similar over the sample period.

[33] Observed bank credit is the sum of FDIC commercial banks' commercial and industrial loans, and construction and development loans. The series is then put into per capita terms. Source: FDIC call reports.

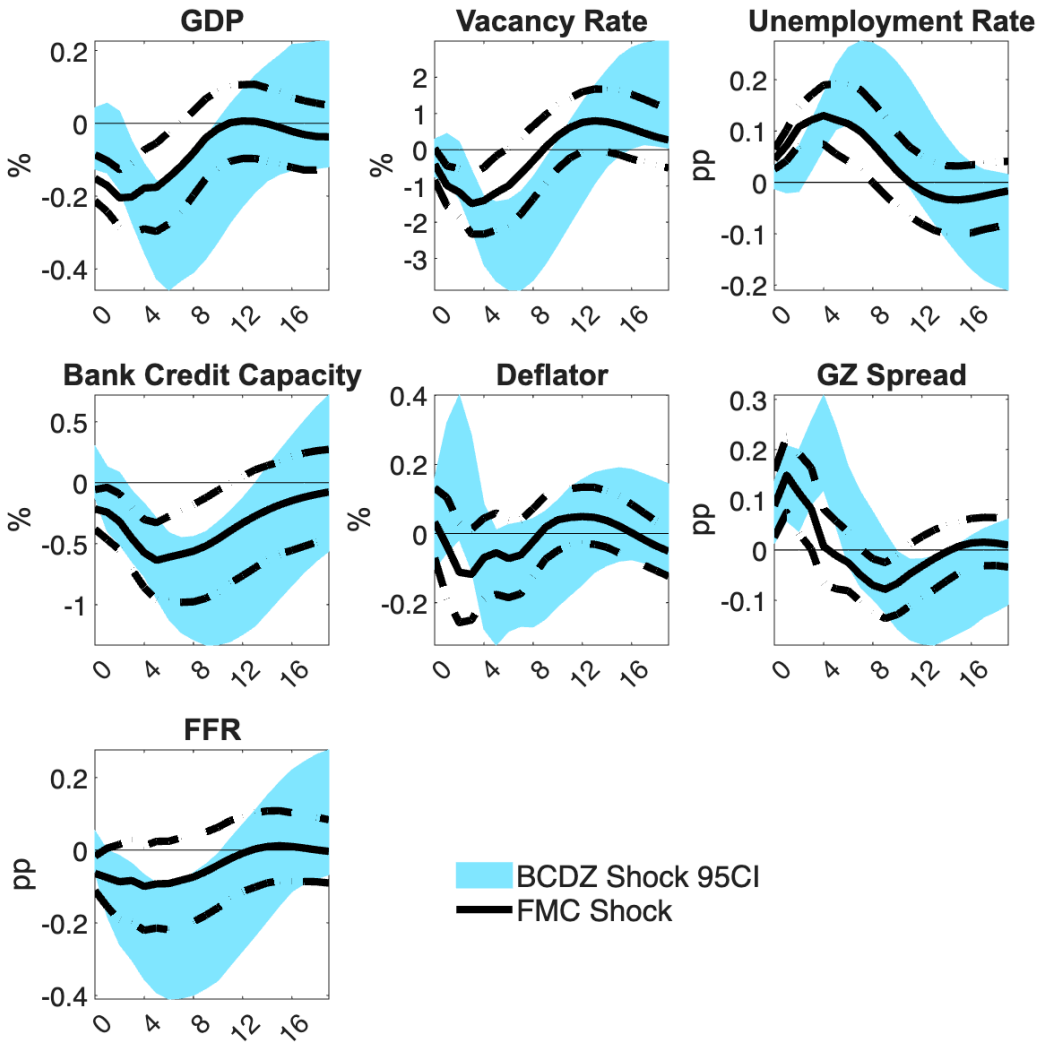

Figure 10: Impulse Responses—BCDZ & FMC shock

Notes: Shaded area denotes the 95% confidence interval of the original VAR model impulse responses estimate in Section 3. The solid lines denotes the VAR model impulse responses when the BCDZ shock is replaced with the model implied FMC shock. Dashed lines denote the respective 95% confidence intervals. See text for further details.

Going further, I ask: *In the context of the VAR model, can the model implied FMC shock generate a VAR response consistent with that of the BCDZ shock*? I replace the BCDZ shock with the model implied FMC shock $\{\hat{\varepsilon}_t^b\}$ in the VAR model. For consistency I also maintain the same sample period as the original VAR model. Looking at Figure 10, it is clear both BCDZ and FMC shocks exhibit a similar model response. Following the FMC shock, however, the variables respond on impact but return to steady state sooner relative to the BCDZ responses.

Alternative FMC Shock Identification: Here I employ an entirely different methodological approach to estimating the FMC shock process. Following Jermann and Quadrini (2012), I use the

model equations to directly isolate the targeted FMC shock process $\{\zeta_{b,t}\}$. In this case the equation of interest is 4 where the FMC shock, in log-deviations from steady state, is

$$\hat{\zeta}_{bt} = -\left[1 + (1+\zeta_b)\rho^d\right]\left(\frac{1+\zeta_b}{\zeta_b}\right) \times \hat{\hat{\rho}}_t^d, \tag{29}$$

where ˆ denotes growth rate. Since the growth in bank's desired reserve ratio is not directly observable, I infer it using corollary 4.1.1. As a result, log-deviations in desired reserve ratio can be written as

$$\hat{\hat{\rho}}_t^d = -\left(\widehat{\int_{i\in\mathscr{M}_t} D_{it}^d \mathrm{d}i} - \hat{D}_t^s\right),$$

where the right-hand side denotes the negative difference between the growth in bank loans to firms, as proxied by $\widehat{\int_{i\in\mathscr{M}_t} D_{it}^d \mathrm{d}i}$, and growth of total reserves held by the bank within period t, $\hat{D}_t^s$.[34] The data used for bank loans is the same series plotted in panel B of Figure 9. As for total reserves held by the bank, I use data of total reserve balances held by depository institutions in master accounts and excess balance accounts at Federal Reserve Banks.[35] After $\hat{\hat{\rho}}_t^d$ is estimated it is straightforward to impute the FMC shock series using equation 29. The alternative FMC shock process estimates imply $\hat{\rho}_b = 0.9799$ with standard error 0.0157, and $\hat{\sigma}_b = 0.4769$. The remaining parameters are kept at their benchmark values.

Figure 11 panel A shows the alternative FMC measure (blue circled line) with the BCDZ shock (solid black line). Again, to maintain consistency between the empirical and model derived bank credit supply shocks, I plot the negative of the alternative FMC shock. One obvious feature of the alternative measure is the sustained downward trend from the 1990s and then the sudden reversal during the financial crisis. Compared to the BCDZ shock, the alternative FMC shock suggests banks contracted credit by very little during the 2001 recession and began contracting credit with about a year lag during the financial crisis.

How does the estimated alternative FMC shock influence the model implied unemployment rate? Panel B of Figure 11 addresses this question. Save for the 1990 recessionary period, the observed (black line) and model implied (blue circled line) unemployment rate exhibit a similar cyclical pattern. Notwithstanding, the sustained credit expansion predicted by the alternative FMC shock prior to the financial crisis has the effect of lowering the unemployment rate below its observed value. Had there been only FMC shocks hitting the U.S. economy, the model predicts that the

[34] While total bank loans in the model are also comprised of bank net worth, as has been shown in the literature bank net worth is 'sticky' quarter-over-quarter (Adrian, Colla, and Song Shin, 2013; Langlais, 2023a). As a result, I assume growth in bank loans is driven by changes in deposit liabilities only.

[35] Sourced via FRED; code BOGMBBM.

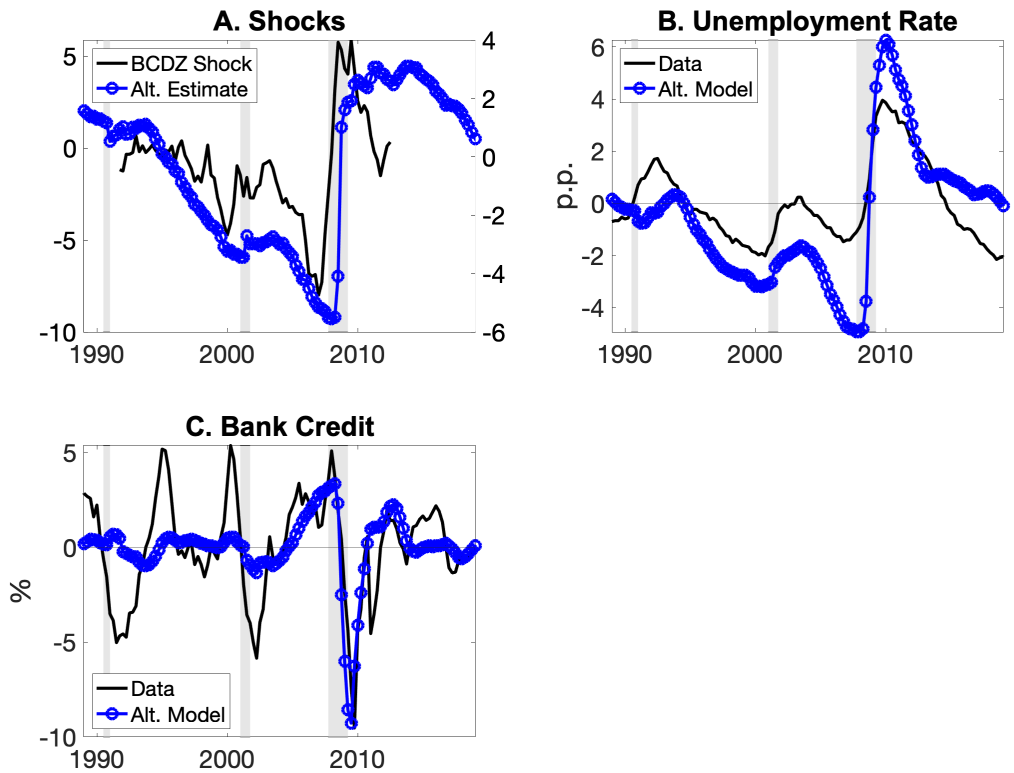

Figure 11: Alternative FMC Shock

Notes: Alternative model parameters are set to the benchmark values except for ρ_b and σ_b which are estimated using the alternative method in the text. Shaded areas denote NBER recession dates.

unemployment rate would have been about 4 percentage points lower than it had been prior to the financial crisis. Additionally, the unemployment rate would have also been 2 percentage points higher towards the end of the crisis. Regarding bank credit (panel C), the model implied series does a poor job of capturing the cyclical pattern of the observed series. During the period surrounding the financial crisis, however, the model implied credit series tracks the observed counterpart closely, especially during the contractionary phase which the benchmark model has trouble replicating.

I cautiously interpret the results presented in this section as evidence that the benchmark model and associated estimated FMC shock do capture movements in the supply of bank credit.

7 Conclusion

This paper asks: Do changes in the supply of bank credit contribute to the observed cyclical pattern of the unemployment rate? If so, how important are bank credit supply shocks at generating fluctuations of the unemployment rate? What are the transmission mechanisms governing such a

relationship? Using a bank credit supply shock estimated in Bassett et al. (2014) I find that bank credit supply shocks account for at least one-third of the volatility in the unemployment rate and vacancy rate at business cycle frequencies. To shed light on the mechanism behind this relationship, I used a model with labor search frictions and nominal rigidities that incorporates a banking sector endowed with a financing through money creation technology allowing it to expand and contract the supply of bank credit independent of the banks' liquidity position. When the bank contracts loans fewer funds are available to firms to purchase inputs and the cost of borrowing rises. Faced with higher borrowing costs, the value of being a matched firm falls and fewer vacancies are posted. As a result, labor market tightness falls and the unemployment rate rises. Both empirical and estimated models suggest that shocks to bank credit supply play a nontrivial role in generating fluctuations in the unemployment rate. Moreover, disruptions to bank credit supply generate fluctuations consistent with a typical business cycle: the unemployment rate and credit spread are countercyclical while investment and labor market tightness are procyclical.

Reasons banks adjust their supply of credit, no doubt, depend on a host of factors. For example, changes in expectations or economic outlook and industry competition have been found to be important determinants of the position of the credit supply curve over the business cycle. While this paper remains agnostic regarding the source of changes affecting the bank credit supply curve, this paper provides evidence that the bank credit supply channel plays an important role in generating cyclical fluctuations in economic activity. As to which sources of variation are driving the results, I leave for future research.

Disclosures

Perplexity, an AI-powered search engine, was used as an aid when completing the literature review. Refine.ink, an AI proofreading tool, was used to check for internal consistency and for clarifying text.

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A Proofs

A.1 Proposition 4.1

Invoking the balance sheet identity *Assets* = *Liabilities* + *Net Worth* on the bank's period t balance sheet (Table 2) yields the following relationship:

$$\rho^s D_t^s + \tilde{\rho}_t^d \int_{i \in \mathscr{M}_t} D_{i,t}^d \mathrm{d}i = D_t^s, \tag{A.1}$$

where the left-hand side denotes total reserves held by the bank. Decomposing the reserves into *required* and *excess* reserves implies

$$\underbrace{\rho^s D_t^s + \tilde{\rho}_t^d \int_{i \in \mathscr{M}_t} D_{i,t}^d \mathrm{d}i}_{\text{total reserves}} = \underbrace{\rho^s D_t^s + \rho^d \int_{i \in \mathscr{M}_t} D_{i,t}^d \mathrm{d}i}_{\text{required reserves}} + \underbrace{E_t}_{\text{excess reserves}} \tag{A.2}$$

which can be rewritten as

$$\tilde{\rho}_t^d = \rho^d + \frac{E_t}{\int_{i \in \mathscr{M}_t} D_{i,t}^d \mathrm{d}i}. \tag{A.3}$$

Noting that $\int_{i \in \mathscr{M}_t} D_{i,t}^d \mathrm{d}i = (\zeta_{b,t} + 1) E_t$ from 2 yields the final result. This concludes the proof.

A.2 Proposition 4.2

See the bank's optimization problem in Section C.2.1 of the Appendix.

A.3 Proposition 4.3

Given $\frac{\partial (J_t^W - J_t^U)}{\partial W_t} = 1$ and $\frac{\partial J_t^F}{\partial W_t} = -(1 + R_t^l)$, the first order condition of the Nash bargaining problem described in the main text can be written as

$$J_t^W - J_t^U = \frac{b}{1-b} \frac{J_t^F}{1 + R_t^l}. \tag{A.4}$$

As in Leduc and Liu (2016), surplus is defined as $S_t \equiv J_t^F + J_t^W - J_t^U$ and plugging it into A.4 yields

$$S_t - J_t^F = \frac{b}{1-b}\frac{J_t^F}{1+R_t^l},$$

then isolating J_t^F yields

$$J_t^F = (1-\omega(R_t^l;b))S_t, \qquad \text{or} \qquad J_t^W - J_t^U = \omega(R_t^l;b)S_t,$$

where $\omega(R_t^l;b) \equiv \frac{b}{b+(1-b)(1+R_t^l)}$. This concludes the proof.

A.4 Proposition 4.4

First, note that given equilibrium condition $J_t^V = 0$, a matched firm's Bellman 21 can be written as

$$J_t^F = Q_t z_t (1-\alpha_f) f(\bar{k}_t) - (1+R_t^l)W_t + \beta \mathbb{E}_t \left\{ \frac{\tilde{\lambda}_{1,t+1}}{\tilde{\lambda}_{1,t}} (1-\delta) J_{t+1}^F \right\}, \tag{A.5}$$

where $\bar{k}_t$ denotes the optimal level of capital per worker chosen by each firm. Under the benchmark model when firms are required to borrow funds to pay their wage bill in advance of production, the job-creation condition—living in the ϑ-w plane—is found by combining A.5 with equilibrium condition 28 in steady state yielding

$$w = \frac{1}{1+R^l}\left[(1-\alpha_f)qz\left(\frac{\varkappa K}{n}\right)^{\alpha_f} - \Phi\frac{\kappa}{p^v(\vartheta)}\right], \tag{A.6}$$

where variables without time subscripts denote the variable's non-stochastic steady-state value and $\Phi \equiv 1-\beta(1-\delta) > 0$.[36] Here q and w are the steady-state values of real output price, Q_t/P_t, and the real wage rate, W_t/P_t, respectively. It is clear to see that the financial frictions introduced in this paper—showing up as $R^l > 0$ in equation A.6—have the effect of shifting the standard job-creation condition downward and rotating it counter-clockwise.

Noting that $S_t = J_t^F / \left(1-\omega(R_t^l|b)\right)$ and $W_t^N/P_t = w$ in steady state, the wage condition—also living in the ϑ-w plane—under the benchmark model is found by combining 26 with the equilibrium condition 28 yielding

$$w = \frac{\psi_L}{\lambda_1} + \mathscr{B} + \frac{1}{1+R^l}\left(\frac{b}{1-b}\right)\left[\Phi + (1-\Phi)p^u(\vartheta)\right]\frac{\kappa}{p^v(\vartheta)}, \tag{A.7}$$

where λ_1 denotes the Lagrange multiplier on the household's asset accumulation equation 9.

[36] Note that since capital markets are competitive, each firm chooses the same capital-labor ratio, implying $\bar{k}_t = \frac{\varkappa K_t}{n_t}$.

Combining the job-creation rule A.6 with the wage rule A.7 yields an expression which, for a given capita-labor ratio and marginal utility of money, pins down the equilibrium market tightness, θ^*:

$$qz(1-\alpha_f)\left(\frac{\varkappa K}{n}\right)^{\alpha_f} - (1+R^l)\left[\frac{\psi_L}{\lambda_1}+\mathscr{B}\right] = \frac{\kappa}{p^v(\vartheta^*)}\left(\Phi + \frac{b}{1-b}\left[\Phi+(1-\Phi)p^u(\vartheta^*)\right]\right). \quad \text{(A.8)}$$

Taking the derivative of equation A.8 with respect to ζ_b, yields

$$\varkappa qz(1-\alpha_f)\alpha_f\left(\frac{\varkappa K}{n}\right)^{\alpha_f-1}\frac{\partial\left(\frac{K}{n}\right)}{\partial\zeta_b} - \left[\frac{\psi_L}{\lambda_1}+\mathscr{B}\right]\frac{\partial R^l}{\partial\zeta_b} + (1+R^l)\frac{\psi_L}{\lambda_1^2}\frac{\partial\lambda_1}{\partial\zeta_b} = \frac{\partial\vartheta^*}{\partial\zeta_b}A_0, \quad \text{(A.9)}$$

where

$$A_0 \equiv \underbrace{\left[\frac{\kappa}{p^v(\vartheta^*)}\frac{b}{1-b}(1-\Phi)p^{u\prime}(\vartheta^*) - \kappa\frac{p^{v\prime}(\vartheta^*)}{p^v(\vartheta^*)^2}\left(\Phi+\frac{b}{1-b}\left[\Phi+(1-\Phi)p^u(\vartheta^*)\right]\right)\right]}_{>0}.$$

Note that the loan interest rate and capital-labor ratio vary inversely to a change in the level of FMC. This can be shown by taking the derivative of the firm's steady-state optimal capital choice—$(1+R^l)r^k = qzf'(k_t)$—with respect to ζ_b,

$$r^k\frac{\partial R^l}{\partial\zeta_b} = \frac{\partial\left(\frac{K}{n}\right)}{\partial\zeta_b}\underbrace{\varkappa qz\alpha_f(\alpha_f-1)\left(\varkappa\frac{K}{n}\right)^{\alpha_f-2}}_{<0},$$

where $r^k = 1/\beta - 1 + \Delta$. Using this last expression to substitute out $\frac{\partial\left(\frac{K}{n}\right)}{\partial\zeta_b}$ from the left-hand side of A.9 yields

$$-\left[\left(\frac{K}{n}\right)r^k + \frac{\psi_L}{\lambda_1} + \mathscr{B}\right]\frac{\partial R^l}{\partial\zeta_b} + (1+R^l)\frac{\psi_L}{\lambda_1^2}\frac{\partial\lambda_1}{\partial\zeta_b} = \frac{\partial\vartheta^*}{\partial\zeta_b}A_0.$$

As a result, since $A_0 > 0$, then if $\frac{\partial R^l}{\partial\zeta_b} < \Xi\frac{\partial\lambda_1}{\partial\zeta_b}$, where

$$\Xi \equiv \frac{(1+R^l)\frac{\psi_L}{\lambda_1^2}}{\left(\frac{K}{n}\right)r^k + \frac{\psi_L}{\lambda_1} + \mathscr{B}} > 0,$$

it must be that $\frac{\partial\vartheta^*}{\partial\zeta_b} > 0$. In words, if the loan rate of interest is sufficiently responsive to the level of FMC, then labor market tightness is positively related to the level of FMC. This is intuitive since a higher level of FMC increases the quantity of funds available to hire labor and lowers the cost of borrowing, thereby allows firms to—optimally—post more vacancies. Note that this relationship

between the cost of borrowing and the labor market tightness is a crucial link between the level of bank FMC and the unemployment rate. To see this I consider the standard relationship arising between labor market tightness and the unemployment rate through the Beveridge curve. Under the labor search timing convention adopted in this paper, the Beveridge curve—living in the ϑ-$\mathscr{U}$ plane—is found by combining equations 18, 19 and 20 yielding

$$\mathscr{U} = \frac{[1 - p^u(\vartheta^*)]\delta}{\delta + (1-\delta)p^u(\vartheta^*)}, \tag{A.10}$$

then taking the derivative of A.10 with respect to ζ_b implies

$$\text{sign}\{\frac{\partial \mathscr{U}}{\partial \zeta_b}\} = \text{sign}\{-p^{u\prime}(\vartheta^*) \times \frac{\partial \vartheta^*}{\partial \zeta_b}\}.$$

Thus, when $\frac{\partial \vartheta^*}{\partial \zeta_b} > 0$ it must be that $\frac{\partial \mathscr{U}}{\partial \zeta_b} < 0$. In words, lowering the level of FMC leads to a higher steady-state unemployment rate. We have established the critical negative relationship between bank's FMC and the unemployment rate. This concludes the proof.

B Data

Table 8: Data Sources

Variable	Description	Source(s)
GDP	GDP in billions of dollars divided by population >= 16yrs and deflated by Implicit Price Deflators for GDP, index 2012=100	Bureau of Economic Analysis: Table 1.1.5. and Table 1.1.9; Federal Reserve Economic Data: CNP16OV
Unemployment Rate	Unemployment Rate, Percent, Seasonally Adjusted, Transformed to quarterly rate	Bureau of Labor Statistics (via Federal Reserve Economic Data: UNRATE)
Vacancy Rate	Job openings rate, Seasonally Adjusted, Transformed to quarterly rate	JOLTS: JTS000000000000000JOR; Prior to December 2010 I use job vacancy series from Leduc and Liu (2020)
Bank Credit Capacity	Gross total loans & leases plus unused loan commitments in millions of dollars divided by population >= 16yrs and deflated by Implicit Price Deflators for GDP, index 2012=100	FDIC Call Reports and Bureau of Economic Analysis: Table 1.1.9; Federal Reserve Economic Data: CNP16OV
Federal Funds Rate	Effective Federal Funds Rate, Percent, Transformed to quarterly rate	Board of Governors of the Federal Reserve System (via Federal Reserve Economic Data: DFF)
Shadow Federal Funds Rate	Estimated shadow Effective Federal Funds Rate when ZLB binds, Percent	Wu and Xia (2016)
GDP Deflator	Index 2012=100	Bureau of Economic Analysis: Table 1.1.9
GZ Spread	Corporate interest rate spread	Gilchrist and Zakrajšek (2012b)
BCDZ shock	Estimated bank lending standards shock	Bassett et al. (2014)
Updated BCDZ shock	Estimated bank lending standards shock over extended sample period	Cavallo et al. (2024)

C Online appendix

C.1 Bank Credit Supply Shock Derived from BVAR with Sign Restrictions

In this section I describe the method used to derive the *sign restriction* ('SR') bank credit supply shock described in the main text.[37] The sign restriction method of identification, instead of assuming the matrix premultiplying the structural shock vector is lower triangular, relies on generating impulse responses—for a given set of parameters—which are consistent with underlying economic theory. For example, consider the VAR model

$$\mathbf{Y}_t = \alpha + \sum_{j=1}^{\rho} \mathbf{A}_j \mathbf{Y}_{t-j} + \mathbf{u}_t, \tag{C.1}$$

where $\mathbf{A}_j$, $j = 1, ..., \rho$ are $n \times n$ matrices of coefficients, and $\mathbf{u}_t = \mathbf{B}\varepsilon_t$, where $\mathbf{B}$ is an $n \times n$ matrix and ε_t is an $n \times 1$ vector of structural shocks with $\mathbb{E}\varepsilon_t = \mathbf{0}$ and $\mathbb{E}\varepsilon_t\varepsilon_t' = I$. The $n \times 1$ vector $\mathbf{u}_t$ are the reduced form shocks such that $\mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_u)$, where $\Sigma_u = B\Sigma_\varepsilon B' = BIB'$.

While equation C.1 can be estimated using standard OLS techniques, under a sign restriction approach it is desirable to use Bayesian vector autoregression (BVAR) methods as it allows simple computation of impulse response function error bands. Moreover, as noted in Uhlig (2005), parameter draws from an estimated posterior distribution are candidate true values and therefore the true impulse responses should not violate the imposed sign restrictions discussed below. As a result, I follow a Bayesian approach which estimates the reduced form coefficients from equation C.1 using an uninformative Normal-Inverse-Wishart prior.

The estimation algorithm is as follows:

1. Estimate the reduced form VAR C.1 using Bayesian methods and obtain S draws from the posterior distribution to get coefficient and covariance estimates $\{\hat{\alpha}^{(s)}, \hat{\mathbf{A}}_1^{(s)}, \hat{\mathbf{A}}_2^{(s)}, \hat{\Sigma}_u^{(s)}\}$ for $s = 1, 2, ..., S$.

2. Obtain the unique Cholesky matrix $C^{(s)}$ for each draw such that $C^{(s)}C^{(s)\prime} = \hat{\Sigma}_u^{(s)}$.

3. Generate a set of random $n \times n$ orthonormal matrices $\{Q^{(k)}\}$ for $k = 1, 2, ..., K$ and set $H^{(s,k)} = C^{(s)}Q^{(k)}$.

4. Check whether the model implied impulse responses for each $\{H^{(s,k)}\}$, $k = 1, 2, ..., K$, satisfy the sign restrictions imposed by the econometrician.

[37] The description here closely follows that in Langlais (2023b). To keep this paper self-contained, however, I reproduce parts of it here with some minor adjustments.

i) if yes, the impulse responses bear a structural interpretation and are saved; set $s = s + 1$ and go to step 2.

ii) if no, set $s = s + 1$ and go to step 2.

Table 9: BVAR Sign Restrictions with Risk Shock

	response variable					
horizon	GDP	Loans	GDP Deflator	GZ[a]	Overnight Rate	$\log(\frac{\text{Uncertainty}}{GZ})^b$
Panel A: loan supply shock						
0	−	−		+	−	−
1	−	−		+	−	−
2	−	−	−	+	−	−
Panel B: aggregate demand shock						
0	−		−	−	−	
1	−		−	−	−	
2	−		−	−	−	
Panel C: aggregate supply shock						
0	−		+		−	
1	−		+		−	
2	−		+		−	
Panel D: monetary policy shock						
0			−		+	
1	−	−	−		+	
2	−	−	−		+	
Panel E: risk shock						
0	−		−	+	−	+
1	−		−	+	−	+
2	−	−	−	+	−	+

Notes: The signs '−' and '+' denote a negative and positive response restriction, respectively. An empty cell denotes no restriction on variable response.
[a] 'GZ' refers to the GZ interest rate spread from Gilchrist and Zakrajšek (2012b).
[b] 'Uncertainty' refers to the total macroeconomic uncertainty measure at the three month horizon from Jurado, Ludvigson, and Ng (2015).

For the analysis below I set $\rho = 2$, $S = 2000$ and $K = 1000$.[38] Following the literature (Mandler and Scharnagl, 2020; Martínez and Rodríguez, 2021; Finck and Rudel, 2022), the variables I choose to include in the BVAR model are real GDP, total bank loan capacity, GDP deflator, interest rate spread and the federal funds rate. Total bank loan capacity is total bank loans plus unused loan commitments and is deflated using the GDP deflator. For the interest rate spread I use the 'GZ' spread from Gilchrist and Zakrajšek (2012b). Except for the GZ spread and the federal funds rate, each variable is transformed into year-over-year growth rates.

[38]This paper uses the MATLAB package developed in Breitenlechner, Geiger, and Sindermann (2019) to perform the estimation algorithm. For further details regarding the algorithm used in this paper, see Rubio-Ramirez, Waggoner, and Zha (2010).

What's more, Langlais (2023a) finds that risk shocks—sometimes referred to as uncertainty shocks—and bank loan supply shock produce a very similar cyclical response in an estimated DSGE model. As a result, I follow Furlanetto, Ravazzolo, and Sarferaz (2019) and include a sixth variable to the model meant to disentangle the model's response to risk and bank loan supply shocks. In particular, I assume that risk shocks have a relatively larger effect on economic uncertainty than they do on interest rate spreads compared to loan supply shocks. As a result, the (log) ratio of economic uncertainty relative to the GZ spread should increase (decrease) following a risk (loan supply) shock. The economic uncertainty measure used here is the 'total macroeconomic uncertainty' measure from Jurado et al. (2015) at the three month horizon. The BVAR sample period is 1985:Q2-2018:Q3.

Table 9 reports the sign restrictions on each of the model variables in response to a loan supply shock (panel A). For expositional purposes, I also list 4 additional shocks: a general aggregate demand shock (panel B), aggregate supply shock (panel C), monetary policy shock (panel D), and a risk shock (panel E). I assume the sign restrictions put on each variable may be imposed for a maximum of 2 periods following each shock. See Langlais (2023b) for a detailed justification of the sign restrictions.

Out of the potential 2 million sets of impulse response functions computed, there were a total of 398 sets of impulse response functions in which the sign restrictions held.

Figure 12 plots the estimated SR shock process of the closest-to-median model. By construction, when the SR shock is positive (negative) bank credit supply is decreasing (increasing). The cyclical pattern of the SR shock is best seen when its cumulative sum is plotted (panel B, blue line). Like the BCDZ shock in Figure 2 of the main text, the SR series exhibits a strong cyclical correlation with the unemployment rate (0.63).

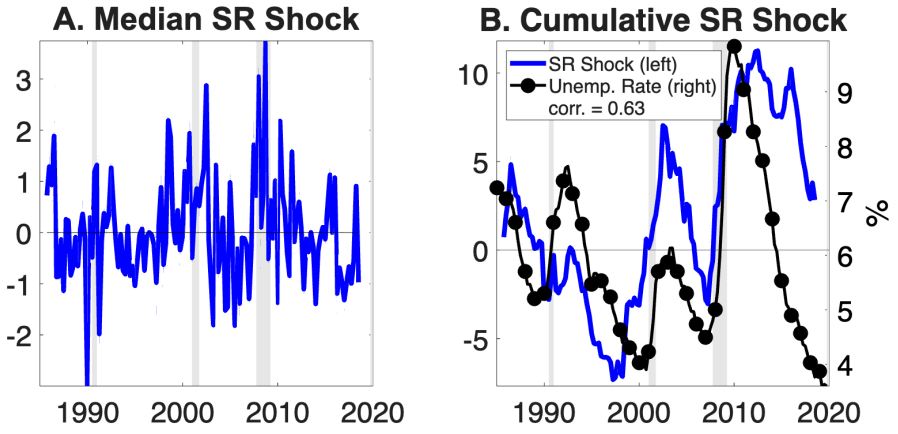

Figure 12: Estimated SR Bank Credit Supply Shock

Notes: Sign restriction (SR) shock process is derived from the closest-to-median model. Correlation coefficient is computed from detrended data from a HP filter with smoothing parameter 1,600. Shaded bars denote NBER recessions.
Source: Bureau of Labor Statistics; author's calculation.

Figure 13 and Table 10 show the IRFs and variance decompositions, respectively, of the SR shock counterpart. When the BCDZ shock is replaced with the estimated SR shock, the estimates shown here are remarkably similar the benchmark VAR results in the main text.

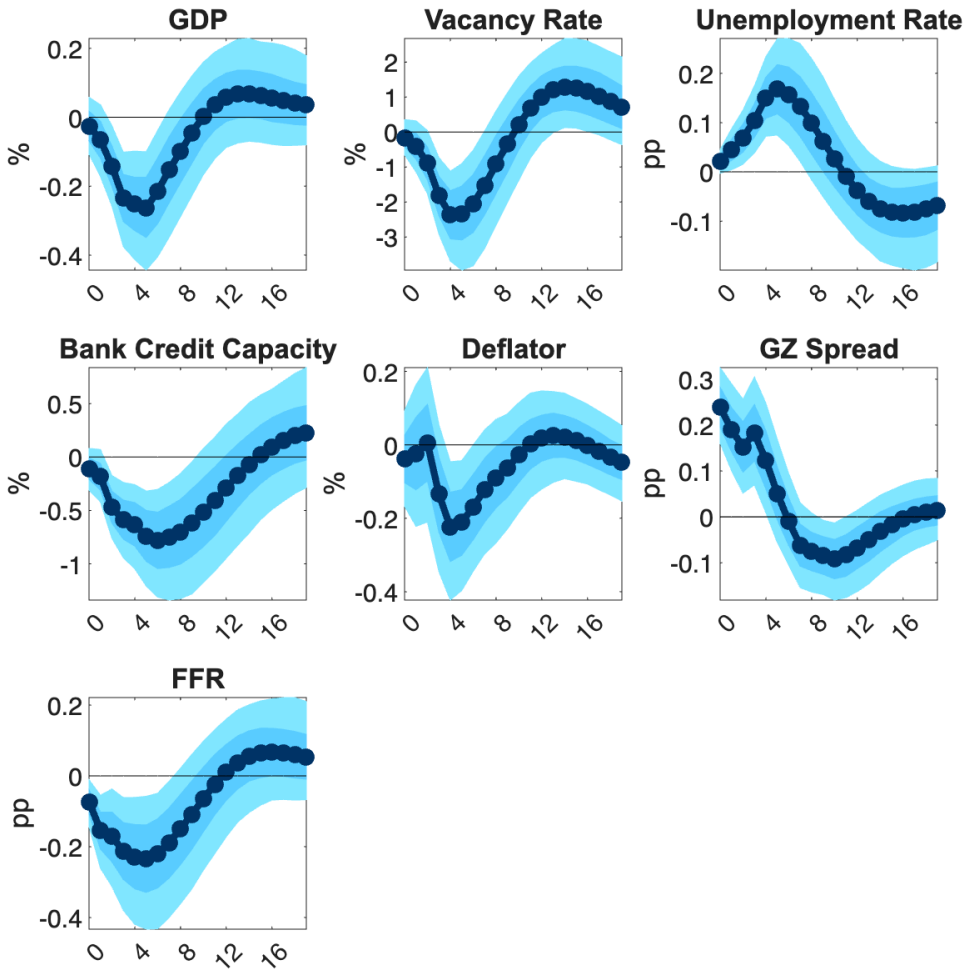

Figure 13: Impulse Responses of Contractionary SR Shock

Notes: Light shaded and dark shaded areas denote 95% and 68% confidence intervals, respectively. Confidence intervals are computed using 2,000 bootstrap draws. See text for further details.

Figure 14 shows the historical decomposition of the vacancy rate and unemployment rate. Again, like the benchmark results, the SR shock plays an important role during the period surrounding the financial crises.

Table 10: VAR Variance Decompositions

horizon	GDP	Vacancy Rate	Unemp. Rate	Bank Credit Capacity	Deflator	GZ Spread	Shadow Federal Funds Rate
$h = 4$	19	17	19	21	5	42	19
	[4–39]	[3–38]	[3–41]	[5–42]	[1–16]	[24–60]	[2–41]
$h = 8$	26	28	30	31	15	36	26
	[7–49]	[7–52]	[8–54]	[9–56]	[3–34]	[19–55]	[5–52]
$h = 16$	23	27	25	24	15	35	23
	[7–43]	[9–48]	[7–47]	[6–50]	[3–33]	[17–53]	[5–48]
$h = 32$	22	26	25	22	16	33	23
	[8–41]	[9–48]	[8–48]	[5–47]	[4–33]	[16–52]	[6–47]

Notes: Each cell entry denotes the percentage contribution attributed to the bank credit supply shock in explaining the volatility of the column variable at horizon h. Entries in square brackets denote the 95% confidence interval based on 2,000 bootstrap draws.

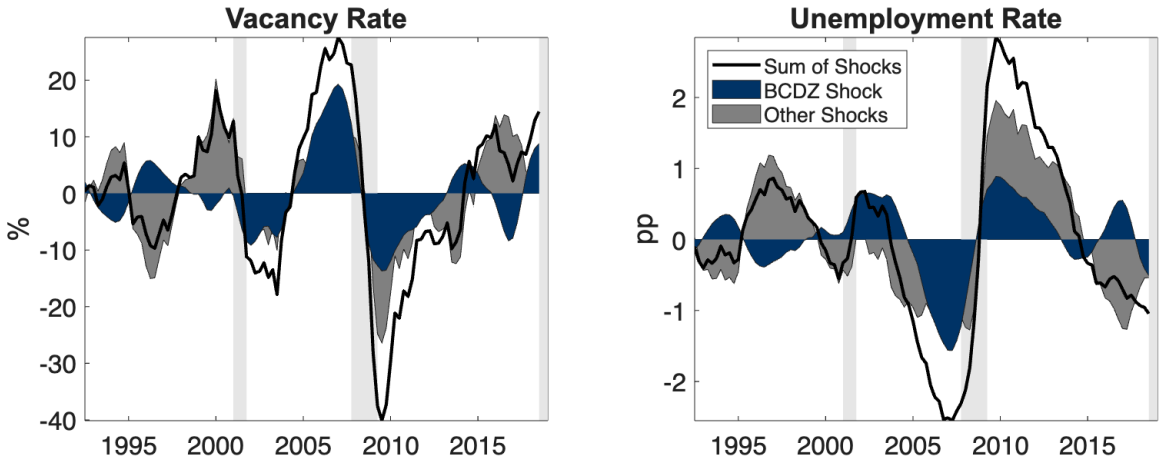

Figure 14: Historical Decomposition of the Vacancy & Unemployment Rate

Notes: Shaded bars denote NBER recessions. Decomposition based on VAR model estimates. See text for further details.

Figure 15 provides and out-of-sample test comparing the benchmark model's implied FMC shock (blue circled line) to the estimated SR shock (black line). From 1990 to 2009, both series show a tight correlation. However, the FMC shock predicts a greater contraction in the supply of loans in the early 2010s.

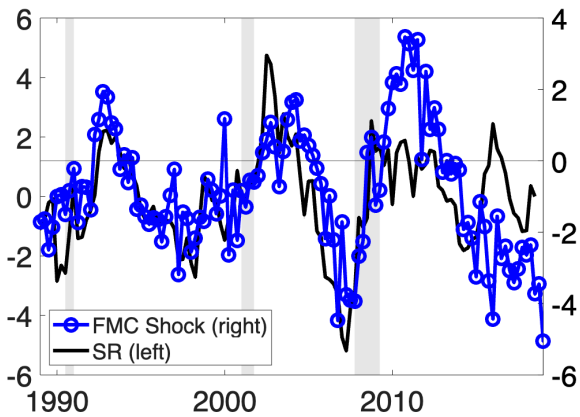

Figure 15: Out of Sample Test—SR Shock versus FMC Shock

Notes: Shaded areas denote NBER recession dates. Sign restriction (SR) shock process (black line) is detrended using the HP filter. SR shock is derived from the closest-to-median model from estimated BVAR model. FMC shock is derived from DSGE model when the estimated parameters are set to their posterior mode.

C.2 Model Optimization Problems

C.2.1 Representative Bank

Although the bank accumulates net worth, the bank's optimization problem is static. Ultimately, in each period the bank's optimization problem is to take previous period's net worth, $NetWorth_t$, interest rates, R_t^d, and R_t^s, and its FMC shock, $\zeta_{b,t}$, as given and choose a level of savings deposits, D_t^s, to issue to the household that maximizes end-of-period profits. To show this I first note that combining 2 and 3, total demand deposits created by the bank can be written as

$$\int_{i \in \mathscr{M}_t} D_{i,t}^d \mathrm{d}i = \frac{1-\rho^s}{\tilde{\rho}_t^d} D_t^s. \tag{C.2}$$

Moreover, using the balance sheet identity, total reserves are simply equal to the amount of savings deposits the bank holds—i.e., D_t^s. Using this fact and using C.2 to substitute out total demand deposits from bank's profit function 6, the bank's optimization problem can be formally written as

$$\max_{D_t^s} \left\{ (1+R_t^l) \left[\frac{1-\rho^s}{\tilde{\rho}_t^d} D_t^s + NetWorth_t \right] + D_t^s - (1+R_t^s) D_t^s - (1+R_t^d) \frac{1-\rho^s}{\tilde{\rho}_t^d} D_t^s - NetWorth_t \right\}.$$

The associated first order condition yields the result from Proposition 4.2.

C.2.2 Representative Household's

Using equation 1 to substitute out M_t, the household's optimization problem can be written as

$$\max_{\{C_t, M_{t+1}^{\text{base}}, D_t^s, K_{t+1}, I_t, \varkappa_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ \log(C_t - hC_{t-1}) - \psi_L n_t + \mathscr{V}\left(\frac{M_t^{\text{base}} - D_t^s}{P_t}\right) \right\},$$

subject to

$$\begin{aligned} M_{t+1}^{\text{base}} \leq\ & W_t n_t + [\varkappa_t r_t^k - a(\varkappa_t)] P_t K_t + P_t \mathscr{B}(1 - n_t) \\ & + (1 + R_{t-1}^s) D_{t-1}^s + M_t^{\text{base}} - D_t^s + X_t + \text{Lump}_t - P_t C_t - P_t I_t, \end{aligned} \tag{C.3}$$

and

$$K_{t+1} = (1 - \Delta) K_t + \left[1 - S\left(\frac{I_t}{I_{t-1}}\right) \right] I_t. \tag{C.4}$$

The associated first order conditions are

$$\begin{aligned}
C_t: \quad & \lambda_{1,t} = \frac{1}{C_t - hC_{t-1}} - \beta \mathbb{E}_t \left\{ \frac{h}{C_{t+1} - hC_t} \right\}, \\
M_{t+1}^{\text{base}}: \quad & \lambda_{1,t} = \beta \mathbb{E}_t \left\{ \frac{1}{\pi_{t+1}} \left[\lambda_{1,t+1} + \mathscr{V}'\left(\frac{M_{t+1}^{\text{base}} - D_{t+1}^s}{P_{t+1}}\right) \right] \right\}, \\
D_t^s: \quad & \lambda_{1,t} = -\mathscr{V}'\left(\frac{M_t^{\text{base}} - D_t^s}{P_t}\right) + \beta \mathbb{E}_t \left\{ \frac{\lambda_{1,t+1}}{\pi_{t+1}} (1 + R_t^s) \right\}, \\
K_{t+1}: \quad & \lambda_{2,t} = \beta \mathbb{E}_t \left\{ \lambda_{1,t+1} r_{t+1}^k + \lambda_{2,t+1}(1 - \Delta) \right\}, \\
I_t: \quad & \lambda_{1,t} = \lambda_{2,t} \left[1 - S\left(\frac{I_t}{I_{t-1}}\right) - S'\left(\frac{I_t}{I_{t-1}}\right) \left(\frac{I_t}{I_{t-1}}\right) \right] + \beta \mathbb{E}_t \left\{ \lambda_{2,t+1} S'\left(\frac{I_{t+1}}{I_t}\right) \left(\frac{I_{t+1}}{I_t}\right)^2 \right\}, \\
\varkappa_t: \quad & r_t^k = a'(\varkappa_t),
\end{aligned}$$

where $\pi_t = P_t / P_{t-1}$. The variable $\lambda_{1,t} \equiv \tilde{\lambda}_{1,t} P_t$, where $\tilde{\lambda}_{1,t}$ and $\lambda_{2,t}$ are the Lagrange multipliers on C.3 and C.4, respectively.

C.2.3 Intermediate-Good Firm

The representative intermediate-good firm optimization problem can be written as

$$\max_{\{n_t, k_t, v_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{\tilde{\lambda}_{1,t}}{\tilde{\lambda}_{1,0}} \left\{ Q_t F(k_t, n_t; z_t) - (1 + R_t^l)[W_t n_t + P_t r_t^k k_t] - P_t \kappa v_t \right\},$$

subject to

$$n_t = (1-\delta)n_{t-1} + p^v(\vartheta_t)v_t, \tag{C.5}$$

where $\beta\frac{\tilde{\lambda}_{1,t+1}}{\tilde{\lambda}_{1,t}}$ denotes the stochastic discount factor, and $\tilde{\lambda}_{1,t}$ denotes the Lagrange multiplier on the household's budget constraint 9. Let J_t^F be the firm's Lagrange multiplier on C.5, then the associated first order conditions are

$$\begin{aligned}
n_t: \quad & J_t^F = Q_t\frac{\partial F(k_t,n_t;z_t)}{\partial n_t} - (1+R_t^l)W_t + \beta\mathbb{E}_t\left\{\frac{\tilde{\lambda}_{1,t+1}}{\tilde{\lambda}_{1,t}}(1-\delta)J_{t+1}^F\right\},\\
k_t: \quad & Q_t\frac{\partial F(k_t,n_t;z_t)}{\partial k_t} = (1+R_t^l)P_t r_t^k,\\
v_t: \quad & P_t\kappa = p^v(\vartheta_t)J_t^F.
\end{aligned}$$

C.3 Optimality Conditions

To solve the model requires to transform each of the nominal variables into real terms by deflating with the final good price level, P_t. Thus, $j_t^F = \frac{J_t^F}{P_t}$, $q_t = \frac{Q_t}{P_t}$. $w_t^N = \frac{W_t^N}{P_t}$, $w_t = \frac{W_t}{P_t}$, $m_{t+1}^b = \frac{M_{t+1}^b}{P_t}$, $d_t^d = \frac{D_t^d}{P_t}$, $e_t = \frac{E_t}{P_t}$. The Lagrange multiplier on the household's asset accumulation constraint is scaled by the price level: $\lambda_{1,t} = P_t\tilde{\lambda}_{1,t}$. The household's savings deposits are scaled by total money base: $d_t^s = \frac{D_t^s}{M_t^b}$. Likewise, the money base growth rate is scaled by total money base: $x_t = \frac{X_t}{M_t^b}$.

The benchmark model can be summarized by 28 endogenous variables:

$$\{C_t,\lambda_{1,t},\lambda_{2,t},\pi_t,\mu_t,p_t^u,p_t^v,n_t,u_t,U_t,Y_t,r_t^k,K_{t+1},I_t,R_t^s,v_t,q_t,j_t^F,w_t^N,w_t,m_{t+1}^b,d_t^s,d_t^d,e_t,R_t^l,R_t^d,networth_{t+1},\varkappa_t\}$$

and 28 equations:

$$\lambda_{1,t} = \frac{1}{C_t - hC_{t-1}} - \beta\mathbb{E}_t\left\{\frac{h}{C_{t+1}-hC_t}\right\} \tag{C.6}$$

$$\lambda_{1,t} = \beta\mathbb{E}_t\left\{\frac{1}{\pi_{t+1}}\left[\lambda_{1,t+1} + \gamma\left(\frac{m_{t+1}^b}{\pi_{t+1}}(1-d_{t+1}^s)\right)^{-\sigma_q}\right]\right\} \tag{C.7}$$

$$\lambda_{1,t} + \gamma\left(\frac{m_t^b}{\pi_t}(1-d_t^s)\right)^{-\sigma_q} = \beta\mathbb{E}_t\left\{\frac{\lambda_{1,t+1}}{\pi_{t+1}}(1+R_t^s)\right\} \tag{C.8}$$

$$\lambda_{2,t} = \beta\mathbb{E}_t\left\{[\lambda_{1,t+1}r_{t+1}^k + \lambda_{2,t+1}(1-\Delta)]\right\} \tag{C.9}$$

$$\lambda_{1,t} = \lambda_{2,t}\left[1 - S(\frac{I_t}{I_{t-1}}) - S'(\frac{I_t}{I_{t-1}})\left(\frac{I_t}{I_{t-1}}\right)\right] + \beta\mathbb{E}_t\left\{\lambda_{2,t+1}S'(\frac{I_{t+1}}{I_t})\left(\frac{I_{t+1}}{I_t}\right)^2\right\} \quad \text{(C.10)}$$

$$K_{t+1} = (1-\Delta)K_t + \left[1 - S(\frac{I_t}{I_{t-1}})\right]I_t \quad \text{(C.11)}$$

$$r_t^k - r^k \exp[\sigma_a(\varkappa_t - 1)] = 0 \quad \text{(C.12)}$$

$$q_t = \frac{1}{\lambda_f} + \xi_p\frac{\lambda_f - 1}{\lambda_f}\left[\pi_t(\pi_t - \pi) - \beta\mathbb{E}_t\left\{\frac{\lambda_{1,t+1}}{\lambda_{1,t}}\frac{Y_{t+1}}{Y_t}\pi_{t+1}(\pi_{t+1} - \pi)\right\}\right] \quad \text{(C.13)}$$

$$\mu_t = \bar{\mu}u_t^{\alpha}v_t^{1-\alpha} \quad \text{(C.14)}$$

$$p_t^u = \frac{\mu_t}{u_t} \quad \text{(C.15)}$$

$$p_t^v = \frac{\mu_t}{v_t} \quad \text{(C.16)}$$

$$n_t = (1-\delta)n_{t-1} + \mu_t \quad \text{(C.17)}$$

$$u_t = 1 - (1-\delta)n_{t-1} \quad \text{(C.18)}$$

$$U_t = 1 - n_t \quad \text{(C.19)}$$

$$Y_t = z_t\left(\frac{\varkappa_t K_t}{n_t}\right)^{\alpha_f} n_t \quad \text{(C.20)}$$

$$C_t + I_t + \kappa v_t + \frac{\xi_p}{2}(\pi_t - \pi)^2 Y_t + a(\varkappa_t)K_t + \phi_b = Y_t \quad \text{(C.21)}$$

$$j_t^F = q_t z_t(1-\alpha_f)\left(\frac{\varkappa_t K_t}{n_t}\right)^{\alpha_f} - (1+R_t^l)w_t + \mathbb{E}_t\left\{\beta\frac{\lambda_{1,t+1}}{\lambda_{1,t}}(1-\delta)j_{t+1}^F\right\} \quad \text{(C.22)}$$

$$q_t z_t \alpha_f\left(\frac{\varkappa_t K_t}{n_t}\right)^{\alpha_f - 1} = (1+R_t^l)r_t^k \quad \text{(C.23)}$$

$$\frac{\kappa}{p_t^v} = j_t^F \quad \text{(C.24)}$$

$$\left(\frac{b}{1-b}\right)\frac{j_t^F}{1+R_t^l} = w_t^N - \frac{\psi_L}{\lambda_{1,t}} - \mathscr{B} + \mathbb{E}_t\left\{\beta\frac{\lambda_{1,t+1}}{\lambda_{1,t}}(1-\delta)\left(1-p_{t+1}^u\right)\left(\frac{b}{1-b}\right)\frac{j_{t+1}^F}{1+R_{t+1}^l}\right\} \quad \text{(C.25)}$$

$$w_t = (w_{t-1})^{\iota}(w_t^N)^{1-\iota} \quad \text{(C.26)}$$

$$R_t^l - R_t^d = \frac{1+(1+\zeta_{b,t})\rho^d}{(1+\zeta_{b,t})(1-\rho^s)}R_t^s, \quad \text{(C.27)}$$

$$d_t^d + \frac{networth_t}{\pi_t} = w_t n_t + r_t^k \varkappa_t K_t \quad \text{(C.28)}$$

$$d_t^d = (1+\zeta_{b,t})e_t \quad \text{(C.29)}$$

$$e_t = (1-\rho^s)\frac{m_t^b}{\pi_t}d_t^s - \rho d_t^d \quad \text{(C.30)}$$

$$networth_{t+1} = (1-\tau)(1+R_t^l)\frac{networth_t}{\pi_t} + \phi_b \quad \text{(C.31)}$$

$$m_{t+1}^{\text{base}} = \frac{m_t^{\text{base}}}{\pi_t}(1+x_t) \tag{C.32}$$

$$R_t^d - R^d = \rho_r(R_{t-1}^d - R^d) + (1-\rho_r)\left[\alpha_\pi \log(\frac{\mathbb{E}_t \pi_{t+1}}{\pi}) + \alpha_y \log(\frac{Y_t}{Y_{t-1}})\right] + \frac{\varepsilon_t^r}{400} \tag{C.33}$$

and 3 exogenous variables $\{z_t, \zeta_{b,t}, x_t\}$ following AR(1) structures

$$\log(\frac{z_t}{z}) = \rho_z \log(\frac{z_{t-1}}{z}) + \varepsilon_t^z, \tag{C.34}$$

$$\log(\frac{\zeta_{b,t}}{\zeta_b}) = \rho_b \log(\frac{\zeta_{b,t-1}}{\zeta_b}) + \varepsilon_t^b, \tag{C.35}$$

$$x_t - x = \rho_x(x_{t-1} - x) + \varepsilon_t^x, \tag{C.36}$$

where $\varepsilon_t^j \sim \mathcal{N}(0, \sigma_j^2)$ with $\sigma_j > 0$ for $j \in \{r, z, b, x\}$.

C.4 Steady State

Although consumption habit parameter h is estimated, for the steady-state analysis I assume $h = 0.641$. Given target π and R^s, combining C.7 and C.8 to find $\beta = \frac{\pi}{\sqrt{1+R^s}}$. From C.9 capital rental rate is found as $r^k = 1/\beta - (1-\Delta)$. From C.12 $\varkappa = 1$. From C.32 we get $1 + x = \pi$. Given target unemployment rate U, $n = 1 - U$. C.17 implies matches $\mu = \delta n$ and from C.18 $u = 1 - (1-\delta)n$, from C.16 vacancies are found as $v = \frac{\mu}{p^v}$ and C.14 implies a matching efficiency of $\bar{\mu} = \frac{\mu}{u^\alpha v^{1-\alpha}}$. From C.15 the probability of a searching worker finding a job is $p^u = \frac{\mu}{u}$. The real price of intermediate-goods from C.13 is found as $q = 1/\lambda_f$. Given the target interest spread, $R^l - R^d$, and savings rate, R^s, from C.27 the exogenous level of FMC is

$$\zeta_b = \left[\frac{R^l - R^d}{R^s}(1-\rho^s) - \rho^d\right]^{-1} - 1.$$

Next, I solve for banking fees, ϕ^b, and the steady-state level of TFP, z, using the following steps:

Step 1: Guess initial value of banking fees ϕ_0^b;

Step 2: Set tolerance ε and find the value of z which solves

$$\left| qz\alpha_f \left(\frac{K}{n}\right)^{\alpha_f - 1} - (1+R^l)r^k \right| < \varepsilon \qquad \text{subject to} \tag{C.37}$$

K and R^l satisfying the following conditions:

i) for given targeted capital-output ratio, $\left(\frac{K}{Y}\right)$, use C.20 to solve for K:

$$K = n \times \left[\left(\frac{K}{Y}\right) z\right]^{\frac{1}{1-\alpha_f}}; \tag{C.38}$$

ii) Find $Y = \left(\frac{K}{Y}\right)^{-1} K$ and from C.11 $I = \Delta K$. Given we target vacancy posting costs of 2% of GDP, $\kappa = 0.02Y$;

iii) Using C.21 $C = Y - I - \kappa v - \phi_0^b$, from C.6 $\lambda = \frac{1}{C}\frac{1-\beta h}{1-h}$, and from C.24 $j^F = \kappa / p^v$;

iv) Following Leduc and Liu (2016), total flow value of non-work $\mathscr{B} + \frac{\psi_L}{\lambda} = 0.75$, thus $\psi_L = \lambda(0.75 - \mathscr{B})$; and

v) Using C.22 and C.25 I solve for R^l and w^N.

Step 3: Once a steady-state TFP, z, and the associated steady-state loan rate of interest, R^l, are found, then for a given targeted interest rate spread, we get $R^d = R^l - [R^l - R^d]$.

Step 4: I solve for d^d, e, m^b, $networth$, and ϕ_1^b using equations C.28, C.31, C.30, C.29 and

$$Lev = \frac{\pi d^d + networth + m^b d^s}{networth}$$

where Lev denotes the bank's asset to net worth ratio and is targeted in the data. If

$$|\phi_1^b - \phi_0^b| < \varepsilon$$

stop; if not, update $\phi_0^b = \phi_1^b$ and return to step 2.

Lastly, using a root-solving method I find parameters $\gamma > 0$ and σ_q which satisfy C.7 and C.8.

C.5 Posterior Densities & Bayesian IRF

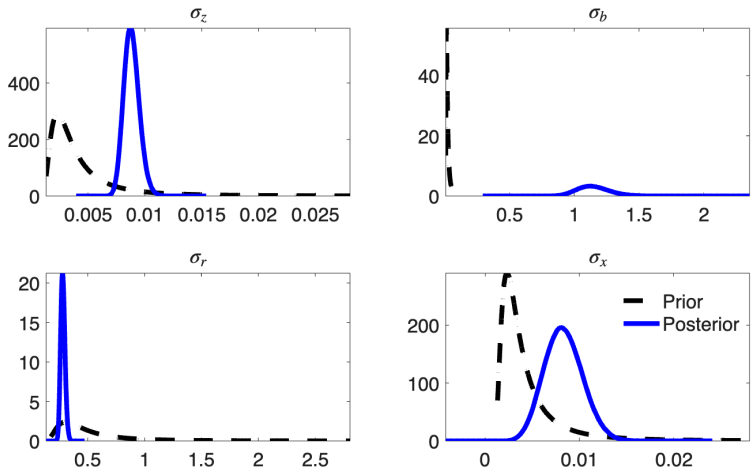

Figure 16: Prior & Posterior Densities: Shocks

Notes: The posterior distribution is obtained from 8,000,000 draws equally distributed across 8 chains of the Metropolis-Hastings algorithm. For each chain the first 20% of the draws are discarded.

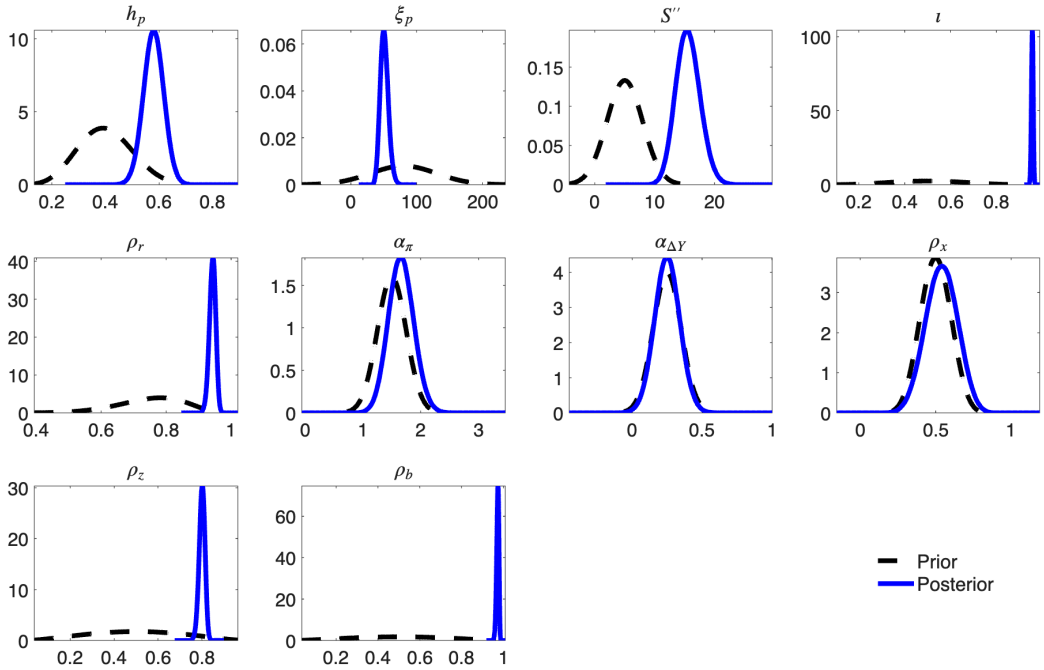

Figure 17: Prior & Posterior Densities: Parameters

Notes: The posterior distribution is obtained from 8,000,000 draws equally distributed across 8 chains of the Metropolis-Hastings algorithm. For each chain the first 20% of the draws are discarded.

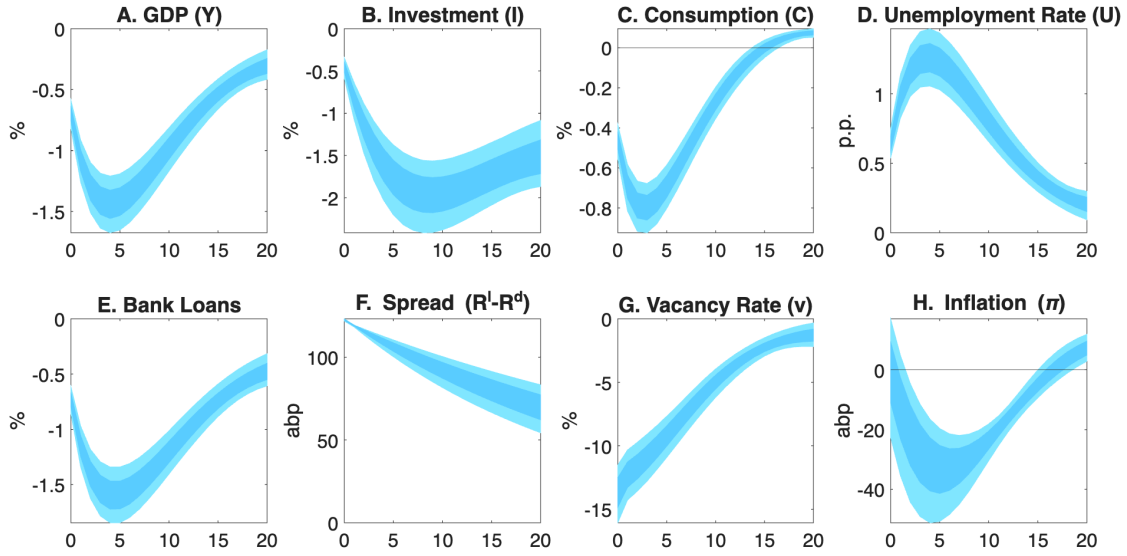

Figure 18: Bayesian Impulse Responses to a FMC Shock

Notes: Bayesian credible intervals are estimated using 10,000 draws from posterior parameter distribution. The light and dark shaded areas denote, respectively, the 95% and 68% equal-tailed credible intervals.

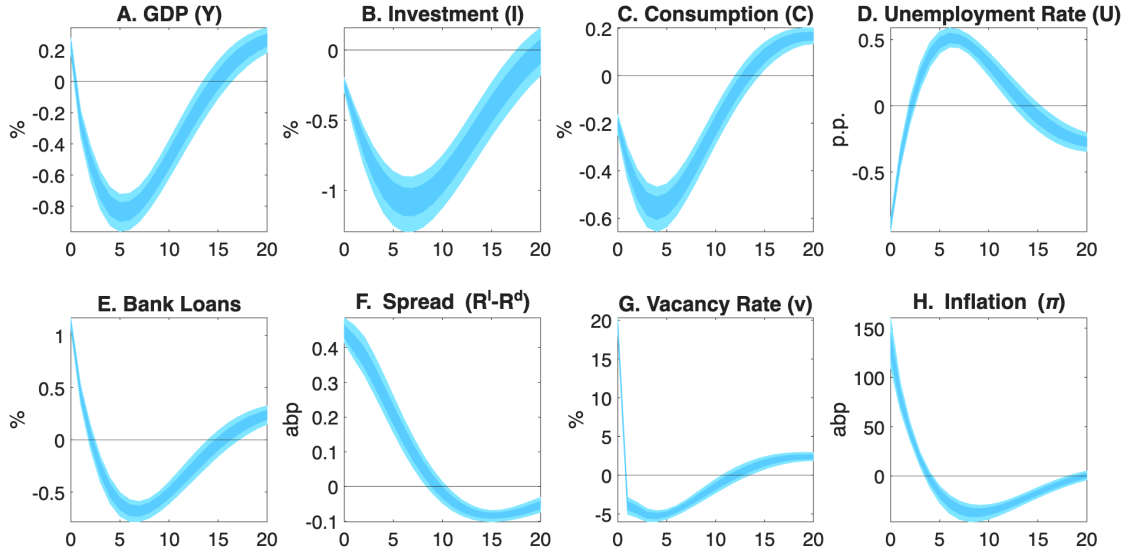

Figure 19: Bayesian Impulse Responses to a TFP Shock

Notes: Bayesian credible intervals are estimated using 10,000 draws from posterior parameter distribution. The light and dark shaded areas denote, respectively, the 95% and 68% equal-tailed credible intervals.

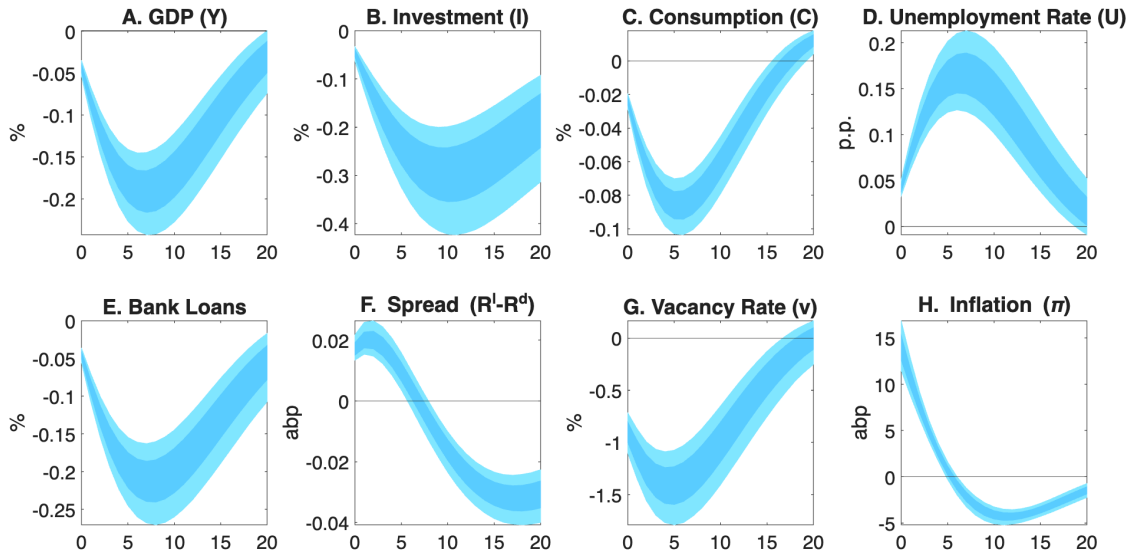

Figure 20: Bayesian Impulse Responses to a Monetary Policy Rate Shock

Notes: Bayesian credible intervals are estimated using 10,000 draws from posterior parameter distribution. The light and dark shaded areas denote, respectively, the 95% and 68% equal-tailed credible intervals.

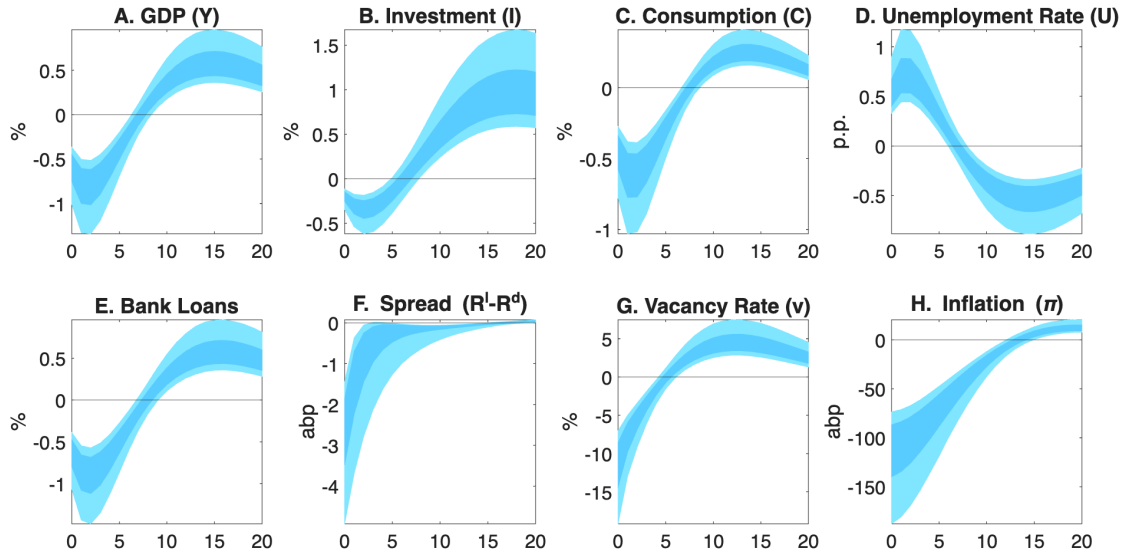

Figure 21: Bayesian Impulse Responses to a Money Base Growth Rate Shock

Notes: Bayesian credible intervals are estimated using 10,000 draws from posterior parameter distribution. The light and dark shaded areas denote, respectively, the 95% and 68% equal-tailed credible intervals.